72nd BPSC PRELIMS

प्रहार Test Series

Test-20/4

FULL LENGTH TEST

Solution

English

Khan Global Studies

Most Trusted Learning Platform

BPSC TEST-20/4 (SOLUTION)

Q 1. Ans. B

Explanation:

- **Annie Besant** founded the **All India Home Rule League** in 1916 and emerged as one of the foremost leaders of the Home Rule Movement. To expand and strengthen the movement in different parts of the country, she visited Bihar in **1918** and mobilised support for the Home Rule League.

Annie Besant (1847–1933) was born on 1 October 1847 in London, England, to an Irish family. A prominent theosophist, social reformer, and nationalist, she represented the Theosophical Society at the World's Parliament of Religions in Chicago in 1893. She played a significant role in India's freedom movement and, in 1917, became the first woman President of the Indian National Congress, presiding over its Calcutta Session. Annie Besant also made notable contributions to education. In 1898, she founded the Central Hindu College at Benares (Varanasi) with the aim of combining India's cultural heritage with modern education. Supported by Theosophists from India and abroad, the institution later became the foundation of the Banaras Hindu University (BHU).

Q 2. Ans. A

Explanation:

- **Patna** was described by the French traveller and merchant **Jean-Baptiste Tavernier** as the **most famous mart in northern India during the middle of the 17th century**. During the Mughal period, Patna emerged as one of the most prosperous commercial centers of eastern and northern India owing to its strategic location on the banks of the Ganga River. The city was renowned for its thriving trade in **saltpetre (an essential ingredient in gunpowder), textiles, silk, opium, sugar, and other commodities**, attracting merchants from across India and Europe.

Q 3. Ans. C

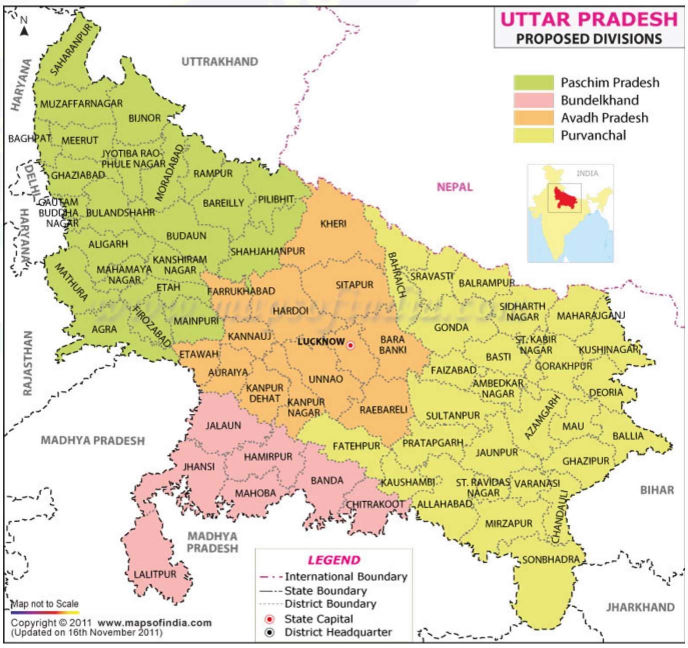
Explanation:

- Bihar is divided into four agro-climatic zones (three major zones, with Zone III further divided into IIIA and IIIB) based on soil characteristics, rainfall, temperature, and terrain.
- Zone I (North-West Alluvial Plains) lies north of the Ganga and includes districts such as Muzaffarpur, Vaishali, and Begusarai. The zone receives moderate to high rainfall and is frequently affected by floods.
- Zone II (North-East Alluvial Plains) receives the highest rainfall in Bihar and is highly flood-prone. Districts such as Katihar, Saharsa, and Madhepura fall within this zone.
- Zone IIIA (South-East Alluvial Plains) is located south of the Ganga and includes Sheikhpura, Munger, and Lakhisarai. The zone is characterised by older alluvial soils and is relatively drought-prone, making it suitable for pulses, oilseeds, and vegetables.
- Zone IIIB (South-West Alluvial Plains) also lies south of the Ganga and includes districts such as Bhojpur, Buxar, Kaimur, and Patna. The region experiences comparatively lower and more variable rainfall.

Q 4. Ans. C

Explanation:

- Bihar shares its western boundary with the state of Uttar Pradesh. A total of seven districts of Uttar Pradesh are situated along the Bihar–Uttar Pradesh interstate boundary. These districts extend from the Nepal border in the north to the Bihar–Jharkhand–Uttar Pradesh tri-junction in the south.
- **The seven Uttar Pradesh districts bordering Bihar are:**
  - ✓ Maharajganj
  - ✓ Kushinagar
  - ✓ Deoria
  - ✓ Ballia
  - ✓ Ghazipur
  - ✓ Chandauli
  - ✓ Sonbhadra



Q 5. Ans. D

Explanation:

The Home Rule League movement in Bihar (1916–1918) constituted an important phase of the Indian national movement aimed at securing self-government (Swaraj) within the British Empire. **The Bihar branch of the League was established at Patna on 16 December 1916, with Maulana Mazharul Haque serving as its provincial president and principal leader.** He was supported by prominent nationalist figures such as Hasan Imam, a former President of the Indian National Congress, and Sachchidananda Sinha. While Patna functioned as the provincial headquarters, the movement expanded through active branches across the state; notably, Babu Janakdhari Prasad established a Home Rule League branch in Muzaffarpur. Another prominent leader, Rai Puran Chand, sought to foster patriotic sentiment by advocating the ceremonial worship of Bharat Mata on Vijayadashami. To further strengthen and reorganise the movement in the province, eminent nationalist leader Dr. M.A. Ansari toured Bihar in 1918, helping to consolidate the League's organisational network and nationalist activities.

Q 6. Ans. D

Explanation:

- On 11 August 1942, during the Quit India Movement, seven students were martyred when the British Indian Military Police opened fire on an unarmed procession of nearly 6,000 students attempting to hoist the national flag atop the Patna Secretariat. The seven martyrs—popularly known as the **"Immortal Seven" (Amar Shaheed), were Umakant Prasad Sinha, Ramanand Singh, Satish Prasad Jha, Jagatpati Kumar, Devipada Choudhry, Rajendra Singh, and Ramgovind Singh. The firing was ordered under the administration of District Magistrate W.G. Archer.** To commemorate their sacrifice, the Shaheed Smarak (Martyrs' Memorial) was erected in front of the Bihar Legislative Assembly and near the Patna Secretariat. Its foundation stone was laid on 15 August 1947 by Bihar's first Governor, Jairamdas Daulatram, in the presence of Sri Krishna Singh, and the bronze sculpture of the seven martyrs was created by renowned sculptor Deviprasad Roychoudhury.

Q 7. Ans. B

Explanation:

- **Dr. Rajendra Prasad** used this statement to highlight Bihar's central role in Indian civilisation. Ancient Bihar was the heartland of major empires such as the **Maurya Empire** and **Gupta Empire**, and was associated with great centres of learning like **Nalanda University** and **Vikramashila University**. The region also played a pivotal role in the rise of Buddhism, Jainism, and several important phases of India's freedom movement.

Q 8. Ans. A

Explanation:

- The **Someshwar Range** forms part of the outer Himalayas (Shiwalik Hills) along the Bihar–Nepal border in **West Champaran district**. A number of river-cut gaps or passes facilitate movement across the range.

Pass	River Associated
Someshwar Pass	Turipani River
Bhikhna Thori Pass	Kudi River
Marwat Pass	Harha River

All three passes are located in the **Someshwar Hills** region of **West Champaran**.

- These passes have been formed through **river erosion**, where streams have cut through the Someshwar Range creating natural corridors.
- **Bhikhna Thori** is an important border location near the **India–Nepal boundary** and has historically served as a transit point.
- The Someshwar Hills constitute the northernmost physiographic division of Bihar and are closely associated with the **Valmiki Tiger Reserve** region.
- The **Harha, Kudi, and Turipani** are small Himalayan foothill streams that have carved these passes through prolonged fluvial action.

Q 9. Ans. C

Explanation:

- The Burhi Gandak River is an important left-bank tributary of the Ganga flowing through northern Bihar. It originates in the Someshwar Hills of West Champaran district near the Indo-Nepal border and follows a highly meandering (sinuous) course across the North Bihar plains before joining the Ganga River near Khagaria.
- Left-Bank Tributaries
  - ✓ Masan
  - ✓ Balor

- ✓ Pandai
  - ✓ Sikta
  - ✓ Tilawe
  - ✓ Tiur
  - ✓ Uria
- Right-Bank Tributaries
  - ✓ Dhanauti
  - ✓ Kohra
  - ✓ Danda
  - ✓ Anjankote
  - ✓ Noon-Balan

Q 10. Ans. C

Explanation:

- **Assertion (A)** is true The **pre-monsoon season** in Bihar extends roughly from **March to mid-June** and is marked by rising temperatures, low humidity during the day, and frequent **convective thunderstorms, dust storms, and squalls**. These local storms, often accompanied by lightning and short-duration heavy rainfall, provide temporary relief from the intense summer heat.
- **Reason (R) is false:** Strong surface heating does **not create atmospheric stability**; rather, it creates **atmospheric instability**. During the pre-monsoon months, intense solar heating warms the land surface, causing hot air to rise rapidly. This rising air generates convection currents, leading to the formation of cumulonimbus clouds, thunderstorms, squalls, and lightning.
- **Why is the Reason Incorrect**
  - ✓ **Strong surface heating** → **Atmospheric instability** →**Convection** →**Thunderstorms and squalls**
  - ✓ The reason incorrectly states that surface heating creates **stability**, which is the opposite of what actually occurs.

Q 11. Ans. A

Explanation:

- **Pair 1 is incorrect: Raja Vishal Ka Garh**, located in the **Vaishali district of Bihar**, is an important archaeological site believed to be the remains of the **assembly hall (parliament) of the Licchavi Republic**, one of the world's earliest republican systems of governance. According to tradition, the fort was constructed by **King Vishal of the Ikshvaku dynasty**, from whom the ancient city of **Vaishali** is said to derive its name.
- **Pair 2 is correct:** The **Gopika Cave (Gopi-ka-Kubha)**, situated on **Nagarjuni Hill** about 35 km north of Gaya, is the largest among the Nagarjuni caves. It was excavated in the **3rd century BCE by the Mauryan ruler Dasaratha**, grandson of Emperor Ashoka. The cave is renowned for its **highly polished Mauryan rock-cut architecture** and inscriptions dedicating it to the **Ajivika sect.**
- **Pair 3 is incorrect:** The **Tomb of Bakhtiyar Khan**, located in the **Chainpur block of Kaimur district**, is a notable **16th-century Indo-Islamic mausoleum** built around **1568 CE**. The octagonal structure serves as the burial monument of **Bakhtiyar Khan**, an Afghan nobleman and father-in-law of **Fateh Khan**, who was married to a daughter of **Sher Shah Suri**, reflecting the region's rich medieval architectural heritage.

Q 12. Ans. B

Explanation:

- **On 13 November 1917, Mahatma Gandhi established the first Basic School (Buniyadi Vidyalaya) in India at Barharwa Lakhansen village** in present-day East Champaran district, Bihar, during the Champaran Satyagraha.
- The school was founded on the principles of "Nai Talim" (Basic Education), which emphasised the integration of academic learning with productive vocational activities such as spinning, weaving, and practical training in hygiene and sanitation. The institution initially functioned from the residence of Babu Shiv Ghulam Lal, a local philanthropist who voluntarily offered his house for the school. The educational activities were managed by Baban Gokhale and Avantikabai Gokhale, an educated couple from Bombay, while Chhotalal Jain supervised vocational instruction. Kasturba Gandhi played a vital role in educating rural women and children, promoting awareness about health, cleanliness, and community sanitation, thereby making the school a pioneering experiment in Gandhian education and rural upliftment.

Q 13. Ans. D

Explanation:

- **Assertion (A) is false:** The **northern Gangetic plain of Bihar is covered by thick alluvial deposits brought by Himalayan rivers such as the Kosi, Gandak, Bagmati, and Mahananda. Hard-rock** mineral resources are generally associated with exposed Precambrian crystalline and metamorphic rocks, which are much more characteristic of the **southern districts of Bihar adjoining the Chotanagpur Plateau margin** (e.g., parts of

Kaimur, Rohtas, Gaya, Nawada, Jamui, Banka, etc.). Therefore, the southern plateau-fringe districts are relatively richer in hard-rock geological formations and associated mineral occurrences than the northern alluvial plain.

- **Reason (R) is true: In the northern Gangetic plain,** thick layers of recent and older alluvium blanket the surface. The ancient crystalline basement lies deep below these deposits and is generally not exposed at the surface. **As a result, the northern plain is dominated by alluvial sediments rather than exposed hard-rock terrain.**

Q 14. Ans. C

Explanation:

- According to the **Census 2011**, the **Urban Male Literacy Rate in Bihar was 82.56%**, the highest among the four literacy categories listed.
- The **Urban Female Literacy Rate** stood at **70.49%.** Although it is significantly higher than rural female literacy.
- The **Rural Male Literacy Rate** in Bihar was **69.67%.** This was lower than urban male literacy but substantially higher than rural female literacy.
- The **Rural Female Literacy Rate** was only **49.00%**, making it the lowest among the four categories.

Q 15. Ans. B

Explanation:

- **Statement 1 is incorrect:** Bihar sends **40 members to the Lok Sabha**, making it one of the largest contributors to the lower house of Parliament. Among these 40 parliamentary constituencies, **6 seats are reserved for candidates belonging to the Scheduled Castes (SCs).** These reservations are provided under constitutional provisions to ensure adequate political representation of the Scheduled Castes in the legislative process.
- **Statement 2 is correct:** The statement is correct because all three constituencies mentioned—**Hajipur, Jamui, and Sasaram**—are included among Bihar's six Scheduled Caste reserved parliamentary constituencies.
- **The 6 SC-reserved Lok Sabha constituencies of Bihar are:**
  - ✓ **Gaya**
  - ✓ **Gopalganj**
  - ✓ **Hajipur**
  - ✓ **Jamui**
  - ✓ **Samastipur**
  - ✓ **Sasaram**

Q 16. Ans. B

Explanation:

- A mycorrhiza is a mutualistic symbiotic association between a fungus and the root system of a living host plant.
- The primary benefit to the plant comes from the vast network of microscopic fungal filaments, called hyphae. These hyphae spread outward and extend far beyond the plant's own standard root zone. They effectively act as an enormous secondary root system, vastly increasing the total physical surface area available to absorb water and minerals from the surrounding soil.
- This extended fungal network is exceptionally efficient at scavenging and mobilizing specific nutrients that are highly immobile in the soil—most notably phosphorus, as well as crucial micronutrients like zinc and copper. The fungus absorbs these elements and delivers them directly into the plant's root cells in exchange for sugars produced by the plant during photosynthesis.

Q 17. Ans. D

Explanation:

- Guard cells are specialised, bean-shaped cells that flank the stomatal pore. Their ability to open and close the pore relies entirely on their unique physical structure and changes in internal water pressure (turgor pressure).
- The cell walls of a guard cell are unevenly constructed. The inner wall (the side directly forming the boundary of the pore) is heavily thickened and rigid. In contrast, the outer wall (facing away from the pore) is thin and highly elastic.
- When environmental conditions are right, water rushes into the guard cells via osmosis. The guard cells swell with this water and become highly pressurised, or turgid.
- As the guard cell swells, the thin, flexible outer wall balloons outward into the surrounding space. Because it is physically connected to the rest of the cell, this outward bulging acts like a lever. It forcefully pulls the thick, rigid inner wall along with it, forcing the inner wall to adopt a curved, crescent shape.
- As both opposing guard cells curve away from each other simultaneously, the central stomatal pore is pulled wide open, allowing for critical gas exchange (intake of carbon dioxide) and transpiration to occur.

Q 18. Ans. C

Explanation:

- Translation is the biological process where the genetic code carried by messenger RNA (mRNA) is decoded to produce a specific sequence of amino acids, which ultimately folds into an active protein or polypeptide chain.
- This process takes place within the cellular structures known as ribosomes. The ribosome reads the mRNA sequence three nucleotide bases at a time (a sequence called a codon). Specialised molecules called transfer RNA (tRNA) recognise these codons and bring the corresponding, specific amino acids to the ribosome. The ribosome then chemically links these amino acids together one by one to build the final polypeptide chain.

Q 19. Ans. B

Explanation:

- Cholesterol is uniquely responsible for maintaining the fluidity and structural integrity of animal cell membranes across a wide range of temperatures. It functions as a bidirectional fluidity buffer within the phospholipid bilayer.
- When temperatures drop, and the membrane threatens to freeze into a rigid, crystalline state, cholesterol molecules wedge themselves between the phospholipid tails. This physical interference prevents the lipid tails from packing tightly together, thereby maintaining membrane fluidity.
- Conversely, when temperatures rise, and the membrane threatens to become too loose and highly permeable, the rigid ring structure of cholesterol restricts the excessive movement of the phospholipid tails, stabilising the membrane and preventing it from falling apart.
- Phosphatidylcholine is the most abundant phospholipid, making up the structural foundation of the membrane, but it does not actively buffer fluidity against temperature changes in the way cholesterol does.
- Glycolipids are lipids with attached carbohydrate chains found on the outer surface of the membrane. Their primary roles are facilitating cellular recognition and assisting in cell-to-cell adhesion, not regulating fluidity.
- Peripheral proteins are temporarily attached to the inner or outer surface of the membrane. They act as enzymes, anchor the cell to its cytoskeleton, or participate in cell signalling pathways, but they do not influence the internal fluidity of the lipid bilayer.

Q 20. Ans. D

Explanation:

- Immunoglobulin G (IgG) is the most abundant type of antibody found in human blood serum, making up approximately 75% to 80% of all circulating antibodies.
- It plays the central role in the body's secondary immune response. Once the immune system has been exposed to a specific pathogen—either through a natural infection or through a vaccination-specialised memory B cells are trained to recognise it. Upon any subsequent exposure to that exact same pathogen, these cells rapidly produce massive quantities of IgG. This mechanism is what provides robust, long-term protective immunity.

Q 21. Ans. B

Explanation:

- The lysogenic phase (or lysogenic cycle) describes the process where a virus successfully infects a host cell, but instead of immediately multiplying and destroying the cell, it physically integrates its own viral genetic material directly into the host cell's DNA.
- Once integrated, this dormant viral DNA is known as a provirus (or prophage in bacteria). As the host cell naturally divides and replicates its own genome, it unknowingly copies the viral DNA along with it. This allows the virus to remain completely hidden and dormant within the host's body for years or even decades without causing any active symptoms. When triggered by environmental stress or immune suppression, the virus can awaken, exit the host genome, and switch to the active lytic phase.
- **Lytic phase:** This is the active reproduction cycle. The virus hijacks the host cell's machinery to build massive amounts of new viral particles, ultimately causing the host cell to burst open (lyse) and die in order to release the new viruses.
- **Viremic phase:** This term refers to the clinical state where active viruses are present and circulating within the host's bloodstream (viremia) to spread to new organs, rather than a cellular cycle.
- **Cytopathic phase:** This refers to the observable structural damage and biochemical changes (cytopathic effects) that occur in a host cell as a direct result of an active viral infection, rather than a state of genetic dormancy.

Q 22. Ans. C

Explanation:

- **Adrenaline** (also known as epinephrine) is the primary hormone secreted by the adrenal medulla, which is the inner portion of the adrenal gland.
- **During an acute stress response, commonly known as the "fight-or-flight" response, the sympathetic nervous system directly stimulates the adrenal medulla to rapidly release adrenaline into the bloodstream. This hormone triggers immediate physiological changes designed to help the body react to a threat:** it increases heart rate, dilates airways for better oxygen intake, redirects blood flow to essential muscles, and triggers a surge of glucose for immediate energy.
- **Cortisol:** While this is widely known as the body's primary "stress hormone," it is secreted by the adrenal cortex (the outer layer of the gland), not the medulla. It is responsible for the body's longer-term stress response, managing metabolism and reducing inflammation over time.
- **Aldosterone:** This hormone is also produced by the adrenal cortex. Its primary biological role is to regulate sodium and potassium levels in the blood, which controls blood volume and blood pressure. It is not an acute stress hormone.
- **Testosterone:** This is an androgenic steroid hormone produced primarily by the testes in males and the ovaries in females (with only trace amounts produced by the adrenal cortex). Its main functions involve the development of male reproductive tissues and promoting secondary sexual characteristics, rather than mediating acute stress.

Q 23. Ans. A

Explanation:

- Cellulose is a naturally occurring polymer. It is an organic polysaccharide consisting of a linear chain of hundreds to thousands of linked D-glucose units. It is the primary structural component of the primary cell wall of green plants, many forms of algae, and oomycetes, making it the most abundant organic polymer on Earth. Because it is synthesised entirely by living organisms in nature, it is classified as a natural polymer.
- **Neoprene:** This is a family of synthetic rubbers that are produced by the polymerisation of chloroprene. It is manufactured industrially and does not occur in nature.
- **Buna-S:** Also known as styrene-butadiene rubber (SBR), this is a synthetic rubber derived from the copolymerization of styrene and butadiene. It is widely used in tyre manufacturing.
- **Bakelite:** This is a synthetic plastic. It is a thermosetting phenol formaldehyde resin, formed from a condensation reaction of phenol with formaldehyde, and holds the distinction of being the first plastic made entirely from synthetic components.

Q 24. Ans. B

Explanation:

- While tin is generally highly resistant to corrosion, which is why it is used to coat and protect the underlying steel or iron of a can from rusting, it is not entirely immune to chemical breakdown.
- When highly acidic foods, such as tomatoes, citrus fruits, or pickled items, are stored in unlined tin cans for prolonged periods, the natural organic acids slowly react with the tin coating. This slow chemical corrosion causes tin ions to dissolve, or leach, directly into the food product.
- Consuming food with excessively high concentrations of leached tin can lead to acute gastrointestinal toxicity, resulting in symptoms like nausea, vomiting, abdominal cramps, and diarrhoea. To prevent this, modern cans designed for acidic foods are typically coated with a protective food-grade epoxy or plastic lacquer that acts as a physical barrier between the acid and the metal.

Q 25. Ans. B

Explanation:

- **Indium Tin Oxide (ITO) is currently the most widely used transparent conducting oxide (TCO) in modern electronics, found in touchscreens, flat-panel displays, organic light-emitting diodes (OLEDs), and solar cells. However, its widespread commercial use faces a critical sustainability challenge:** the scarcity and cost of Indium.
- Indium is a rare element, primarily obtained as a minor byproduct of zinc ore processing rather than being mined directly. Because its supply is tightly linked to the production of other metals, it is highly vulnerable to supply chain disruptions and extreme price volatility. As global demand for consumer electronics grows, the reliance on a scarce material like Indium makes manufacturing economically risky and expensive.
- **To overcome these limitations, scientists are**

developing alternative transparent conductors such as:

- ✓ **Fluorine-doped Tin Oxide (FTO) and Aluminum-doped Zinc Oxide (AZO):** These utilise much more abundant and cheaper materials (zinc, aluminium, fluorine).
- ✓ **Silver Nanowire (AgNW) networks:** Highly flexible and highly conductive.
- ✓ **Graphene and Carbon Nanotubes:** Carbon-based materials offering excellent mechanical flexibility.
- ✓ **Conductive Polymers (e.g., PEDOT:**PSS): Ideal for low-cost, flexible, and printable electronics.

Q 26. Ans. D

Explanation:

- Fluorescence is an almost instantaneous process. The material absorbs light and immediately re-emits it. Once the light source is removed, the glow stops within fractions of a microsecond.
- Phosphorescence, on the other hand, acts as a "store" of light. The material continues to emit a visible glow for a noticeable period, ranging from milliseconds to several hours, after the excitation source is removed. This is the mechanism behind common "glow-in-the-dark" materials.
- While both are forms of photoluminescence, the delay in phosphorescence is due to quantum mechanics. Fluorescence involves a rapid, "spin-allowed" electron transition (from a singlet excited state). Phosphorescence involves a "spin-forbidden" transition (from a triplet excited state); because the electron must physically flip its spin state to return to its resting ground state, the energy release is bottlenecked and highly delayed.

Q 27. Ans. D

Explanation:

- **A fuse is a critical safety device designed to protect electrical circuits from damage caused by excessive current (overcurrent) or short circuits. To function properly, the material used for the fuse wire must possess two specific properties:**

1. Low Melting Point: The primary function of a fuse is to melt and physically break the circuit when too much current flows through it. A low melting point ensures that the wire will melt quickly as soon as the safe current threshold is exceeded, cutting off the power before the excessive current can damage connected appliances or start a fire. Common fuse materials include alloys of lead and tin.
2. High Resistivity: According to Joule's law of heating, the heat generated in a conductor is directly proportional to its electrical resistance. Because resistance is dependent on the material's resistivity, a high resistivity ensures that the fuse wire heats up much faster than the standard copper wiring in the rest of the circuit. This guarantees the fuse is the localised point that melts.

Q 28. Ans. B

Explanation:

- A fuse is a critical safety device that is always connected in series with the live wire of an electrical circuit.
- When an electrical fault occurs (like a short circuit or an overload), and the current exceeds a safe limit, the high resistance and low melting point of the fuse wire cause it to heat up and melt.
- Once the fuse wire melts, it physically breaks the electrical connection, creating what is known as an open circuit. Because there is no longer a continuous, unbroken path for the electrons to flow through, the electrical current drops immediately to zero, and the bulb goes off.

Q 29. Ans. C

Explanation:

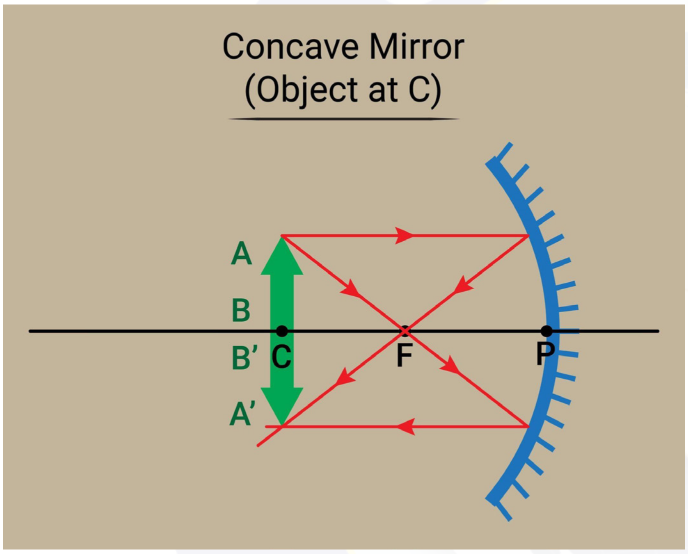
- **When resistors are connected in parallel, you use this formula to find their combined (equivalent) resistance:**
  - ✓ 1/Total Resistance = (1 / R1) + (1 / R2)
- **Plug in the two resistor values (6 and 3):**
  - ✓ 1/Total Resistance = 1/6 + 1/3
  - ✓ 1/Total Resistance = 1/6 + 2/6
  - ✓ 1/Total Resistance = 3/6
  - ✓ 1/Total Resistance = 1/2
- By flipping the fraction, we find that the total resistance of the circuit is 2 ohms.
- **Now, use Ohm's law (Voltage = Current x Resistance) and rearrange it to solve for current:**
  - ✓ Current = Voltage/Total Resistance
- **Plug in the battery voltage (12 V) and the total resistance you just found (2 ohms):**
  - ✓ Current = 12/2
  - ✓ Current = 6 A
- The total current drawn from the battery is 6 Amperes.

Q 30. Ans. B

Explanation:

- In a concave mirror, the point located at exactly twice the focal length is called the Center of Curvature (often labelled as C).

- **When an object is placed exactly at this centre point, the light rays bounce off the mirror and meet back at that exact same location. Because of how the light reflects, the resulting image has three specific features:**
  - ✓ **Real:** The light rays actually meet, meaning you could project this image onto a physical piece of paper or a screen.
  - ✓ **Inverted:** The image is flipped upside down compared to the original object.
  - ✓ **Same size:** The image is exactly the same height and width as the original object.



Q 31. Ans. C

Explanation:

- In physics, work is only done when a force causes an object to move in the direction of that force.
- In this scenario, the force of gravity is pulling the box straight down. However, the box is being carried horizontally (straight across). This means the angle between the downward pull of gravity and the forward movement of the box is exactly 90 degrees.
- Because the movement is completely perpendicular to the force of gravity, no energy is transferred in the vertical direction. Even though it might feel tiring to hold a 20 kg box, from a strict physics perspective, the work done against gravity while moving horizontally at a constant height is exactly 0 Joules.

Q 32. Ans. C

Explanation:

- **As a pendulum swings back and forth, it constantly trades energy between two forms:** potential energy (energy of height) and kinetic energy (energy of motion).
- At the highest points of its swing on either side, the pendulum momentarily stops before reversing direction. Because it is at its highest point, its potential energy is at its maximum. Because it is momentarily stopped, its kinetic energy is zero.
- As gravity pulls it downward toward the centre, it loses height but gains speed. The potential energy is actively converted into kinetic energy.
- When the pendulum bob reaches its absolute lowest point, it is moving at its fastest possible speed, meaning its kinetic energy is at its absolute maximum. Because it is at its lowest height, its potential energy is at its absolute minimum.

Q 33. Ans. B

Explanation:

- When a sound wave travels from one medium to another (like from air into water), its speed changes. The speed of sound depends entirely on the physical properties of the material it is travelling through, specifically its elasticity and density. Because water is much more difficult to compress than air, sound travels significantly faster in water (about 1480 meters per second) than it does in air (about 343 meters per second).
- The frequency of the sound wave, however, is determined solely by the original source that created the sound. Once the wave is created, its frequency remains absolutely constant, no matter what material it passes through. Because the time period is just the inverse of frequency, it also remains constant.
- According to the wave equation (Speed = Frequency x Wavelength), if the frequency stays exactly the same but the speed increases, the wavelength must also increase to balance the equation. Therefore, both the speed and the wavelength change as the sound wave enters the water.

Q 34. Ans. B

Explanation:

- **Mass of water:**
  - ✓ Since 1 litre of water has a mass of 1 kg, 500 litres = 500 kg.
- **Work done in lifting:**
  - ✓ **Using $g = 10\ m/s^2$:**
    - ▪ $W = mgh = 500 \times 10 \times 10 = 50{,}000$ J
- Time.
  - ✓ The water is lifted per minute, so t = 60 seconds.

- Power:
  - ✓ P = W/t = 50,000 / 60 = 833.3 W

Q 35. Ans. B

Explanation:

- **This problem is solved using Newton's second law of motion, which states that the acceleration of a body is directly proportional to the net force acting on it and inversely proportional to its mass, expressed as:**
  - ✓ a = F_net/m.
  - ✓ The crucial point is that acceleration is governed by the net (resultant) force, not merely the force applied by the engine.
- In this situation, two forces act on the car along the horizontal direction. The engine exerts a driving force of 4500 N that pushes the car forward, while the force of friction, measuring 1500 N, acts in the opposite direction to oppose the car's motion.
  - ✓ Since these two forces act in opposite directions, the net force is obtained by subtracting the opposing frictional force from the forward driving force.
- **Therefore, the net force acting on the car is:**
  - ✓ F_net = 4500 − 1500 = 3000 N, directed forward in the direction of motion.
- **Substituting this net force and the car's mass of 1500 kg into Newton's second law gives:**
  - ✓ a = 3000/1500 = 2 m/s^2. The positive value confirms that the car accelerates in the direction of the engine's driving force.
- Hence, the acceleration of the car is 2 m/s^2.

Q 36. Ans. D

Explanation:

- According to Newton's First Law of Motion, an object will stay in motion at a constant speed unless acted upon by an unbalanced force.
- While the block was moving at a constant velocity, the forward push (applied force) was perfectly balanced by the backward drag of the rough surface (kinetic friction).
- The moment the forward push is removed, that balance is broken. The backward pull of kinetic friction becomes the only horizontal force acting on the block. Because this unbalanced friction force pushes against the direction of movement, it forces the block to slow down (decelerate) until it completely stops.

Q 37. Ans. A

Explanation:

- Phase 2 of India's Ballistic Missile Defence (BMD) programme consists of two interceptor missiles, the AD-1 and the AD-2, designed to neutralise more advanced, longer-range ballistic missile threats than the Phase 1 system.
- The AD-1 is a long-range interceptor designed for both endo-atmospheric and low exo-atmospheric interception of long-range ballistic missiles (and even aircraft); it is propelled by a two-stage solid motor. The AD-2 is primarily an exo-atmospheric interceptor optimised to destroy incoming missiles at high altitudes during their mid-course flight phase, outside the Earth's atmosphere.
- DRDO conducted the maiden flight test of the Phase-2 AD-1 interceptor on 2 November 2022 from APJ Abdul Kalam Island off the Odisha coast, and an end-to-end Phase-2 BMD system test on 24 July 2024. The programme was then completed with successful tests of both the AD-1 and AD-2 interceptors conducted on 10–11 June 2026 from the Integrated Test Range off the Odisha coast, in which the interceptors engaged and destroyed targets in both endo-atmospheric and exo-atmospheric flight. These tests put India among the small group of nations with BMD capability to engage ballistic missiles up to the ICBM class.

Q 38. Ans. A

Explanation:

- Vyommitra (derived from the Sanskrit words "Vyoma" meaning space, and "Mitra" meaning friend) is a female-looking half-humanoid robot developed by the Indian Space Research Organisation (ISRO).
- Before India sends human astronauts into space as part of the Gaganyaan programme, ISRO will launch Vyommitra aboard the uncrewed Gaganyaan G-1 test flight. Because she is a "half-humanoid" (having a head, torso, and arms, but no legs), she is specifically designed to remain seated and simulate a human astronaut's physical functions.
- Her primary role during the uncrewed flight is to test the Environmental Control and Life Support Systems (ECLSS), operate switch panels, monitor internal cabin conditions like air pressure, and transmit this critical safety data back to Earth. This robotic testing is a crucial step to verify that the crew module is fully safe and habitable before actual astronauts fly in the subsequent crewed missions.

Q 39. Ans. B

Explanation:

- The 'Shakti' advanced Electronic Warfare (EW) suite is an indigenous system designed and developed by the Defence Research and Development Organisation (DRDO) for the Indian Navy. Its primary function is the interception, detection, classification, identification, and jamming of conventional and modern radars.
- **By providing an electronic layer of defense, Shakti is designed to ensure survivability and electronic dominance in the maritime battlefield. It protects naval platforms against anti-ship missiles by actively disrupting and deceiving the enemy's radar tracking and communication networks. (Note:** While "Operation Sindoor in 2025" appears to be a hypothetical scenario or mock-test context, the real-world function of the DRDO Shakti system remains strictly electronic warfare and radar jamming).

Q 40. Ans. D

Explanation:

- Project Kusha is an ambitious, indigenous defense initiative led by the Defence Research and Development Organisation (DRDO) to build a Long-Range Surface-to-Air Missile (LR-SAM) system for the Indian Air Force. Often compared to the Russian S-400 or the Israeli Iron Dome in terms of its strategic capability, the system is designed to provide a highly comprehensive air defense shield.
- It features advanced interception capabilities designed to detect and destroy a wide variety of aerial threats at ranges up to 350 kilometers. This includes enemy aircraft (even stealth fighters), cruise missiles, drones (UAVs), and precision-guided munitions. Because statements A, B, and C accurately describe the project's origin, developer, and purpose, "More than one of the above" is the correct choice.

Q 41. Ans. D

Explanation:

- QRSAM is specifically designed for the Indian Army to provide on-the-move, mobile air defense cover to advancing armored columns and strike formations. The entire weapon system is configured on mobile vehicles, allowing it to operate continuously while on the move.
- The system was indigenously designed and developed by the Defence Research and Development Organisation (DRDO) in close collaboration with defense public sector undertakings like Bharat Electronics Limited (BEL), which heavily contributed to the radar and command-and-control systems, and Bharat Dynamics Limited (BDL).
- The QRSAM system features advanced Active Electronically Scanned Array (AESA) radar and a state-of-the-art active radio frequency (RF) seeker. This advanced tracking technology allows the system to search, track, and engage multiple aerial targets at the exact same time.

Q 42. Ans. C

Explanation:

- **Quantum tunneling** is the quantum-mechanical phenomenon in which a particle passes through a potential energy barrier even though its kinetic energy is less than the height of the barrier, something that is strictly impossible under classical physics.
- In classical mechanics, a particle that lacks sufficient energy to surmount a barrier is simply reflected back; it can never appear on the other side. In quantum mechanics, however, a particle is described by a wavefunction, and this wavefunction does not abruptly drop to zero at the barrier. Instead, it decays exponentially inside the barrier and retains a small but non-zero amplitude on the far side. This means there is a finite probability that the particle will be found beyond the barrier, effectively having "tunnelled" through it. The thinner and lower the barrier, and the lighter the particle, the greater the probability of tunneling.
- **Quantum entanglement:** This refers to a phenomenon where two or more particles become fundamentally linked. The quantum state of one entangled particle instantly dictates the state of the other, regardless of the physical distance separating them (what Einstein famously called "spooky action at a distance").
- **Quantum superposition:** This is the principle that a quantum system can simultaneously exist in multiple different states, positions, or configurations at the same time until it is physically measured or observed, at which point it "collapses" into a single state.
- **Quantum decoherence:** This describes the process by which a quantum system interacts with its macroscopic environment. This interaction causes

the system to leak information, destroying its delicate quantum properties (like superposition) and forcing it to behave according to classical physics.

Q 43. Ans. B

Explanation:

- **Natural rubber in its raw state has several limitations:** it becomes soft and sticky when warm, and brittle when cold. To make it usable for everyday applications like car tires and shoe soles, it undergoes a chemical process called vulcanisation, famously developed by Charles Goodyear.
- During vulcanisation, raw rubber is heated in the presence of sulphur. The sulphur atoms form strong chemical bonds, or "cross-links," between the long, parallel chains of the rubber polymer (polyisoprene). You can think of these sulphur cross-links like the rungs of a ladder holding the side rails together.
- This three-dimensional network prevents the polymer chains from slipping past each other when stretched, drastically improving the rubber's elasticity, tensile strength, and resistance to temperature changes.

Q 44. Ans. B

Explanation:

- A Light Emitting Diode (LED) is a specialized semiconductor device that produces light through a process called electroluminescence.
- **Inside an LED, there is a junction between two types of semiconductor materials:** an n-type (which has extra electrons) and a p-type (which has extra "holes," or missing electrons). When an electrical current is applied to the diode, it forces the extra electrons and holes to meet at the junction. As the electrons drop into these holes, they transition to a lower energy state. The excess energy they lose during this drop is released directly as tiny packets of light called photons.

Q 45. Ans. C

Explanation:

- For the human ear to distinguish two separate sounds (the original sound and the echo), there must be a time interval of at least 0.1 seconds between them. This biological delay is known as the persistence of hearing.
- **Assuming the average speed of sound in air is about 340 to 344 meters per second at standard temperatures, we can calculate the total distance the sound must travel during that 0.1-second window using the basic formula:** Distance = Speed x Time.
- Total distance = 340 m/s x 0.1 s = 34 meters.
- Because an echo requires the sound to travel to the reflecting surface and then bounce all the way back to your ear, the total distance of 34 meters is a round trip. To find the minimum distance to the surface itself, you simply divide that total round-trip distance by 2.
- 34 meters/2 = **17 meters.**

Q 46. Ans. A

Explanation:

- **Statement 1 is correct:** Magadha's rise was aided by its proximity to the rich iron ore deposits of present-day south Bihar and the Chotanagpur Plateau. The availability of iron facilitated the manufacture of agricultural implements and superior weapons, strengthening both its economy and military power.
- **Statement 2 is correct:** Magadha is regarded as the first major state in ancient India to employ war elephants extensively in organised warfare. The forests of the region provided an abundant supply of elephants, which became a decisive military advantage in its expansion.
- **Statement 3 is incorrect:** Early Magadhan rulers did not exclusively rely on Vedic rituals for legitimacy. Several rulers patronised or tolerated emerging heterodox traditions such as Buddhism and Jainism. For instance, Bimbisara and Ajatashatru are associated with both the Buddha and Mahavira in Buddhist and Jain traditions.

Q 47. Ans. D

Explanation:

- The Satavahanas are credited with initiating the large-scale practice of granting tax-free lands (Agraharas) to Brahmans and Buddhist monks. These grants marked an important development in the agrarian and administrative history of ancient India.
  - ✓ The donated lands were often exempt from various taxes and sometimes carried administrative and fiscal privileges.
  - ✓ Such grants helped in the spread of Brahmanical influence, supported Buddhist monastic establishments, and facilitated the expansion of agriculture into new areas.
  - ✓ The practice became much more widespread during the post-Gupta period, but its large-scale beginnings are generally associated with the Satavahanas.

Q 48. Ans. A

Explanation:

- **Statement 1 is correct:** During the Sangam Age, the Tamil kingdoms maintained extensive maritime trade with the Roman Empire. Roman merchants imported commodities such as pepper, fine textiles, pearls, ivory, and precious stones, while large quantities of Roman gold and silver coins entered South India in return. Numerous Roman coin hoards discovered in Tamil Nadu attest to this flourishing trade.
- **Statement 2 is incorrect:** The term Yavanapriya literally means "dear to the Yavanas (Greeks/ Romans)" and was used to refer to pepper, one of the most sought-after exports of the Tamil region. It did not denote imported Chinese silk.

Q 49. Ans. C

Explanation:

- The Charvaka (Lokayata) School was the only strictly materialistic and atheistic school among the major systems of ancient Indian philosophy.
  - ✓ **It held that the physical universe is composed solely of four material elements:** earth, water, fire, and air. It rejected the existence of a fifth element (akasha) as an independent reality.
  - ✓ The school denied the authority of the Vedas and rejected divine revelation as a valid source of knowledge.
  - ✓ It refuted the concepts of soul (atman), karma, rebirth, heaven, and hell, maintaining that death marks the end of individual existence.
  - ✓ Charvakas accepted direct perception (pratyaksha) as the only reliable means of knowledge, while rejecting inference and scriptural testimony as independent sources of truth.

Q 50. Ans. A

Explanation:

- The Harappans procured several raw materials through long-distance trade networks.
  - ✓ Lapis Lazuli → Shortughai (Northern Afghanistan)
- Shortughai was a Harappan trading outpost near the lapis lazuli mines of Badakhshan in Afghanistan.
- Lapis lazuli was a highly prized semi-precious stone used in beads and ornaments.
- **Additional fact:**
  - ✓ Lothal was a major port town, not a source of jade.
  - ✓ Badakhshan is famous for lapis lazuli, not copper.
  - ✓ Harappan copper was mainly obtained from regions such as Khetri in Rajasthan.
  - ✓ Khetri is known for copper deposits.
  - ✓ Steatite (soapstone) came from other regions and was widely used for seals and beads.

Q 51. Ans. B

Explanation:

- The Vajapeya ("drink of strength" or "drink of vigour") was one of the major royal sacrifices of the Later Vedic period, and its central and most distinctive feature was a ritual chariot race in which the sacrificer-king's chariot was made to win, symbolising the renewal and reinvigoration of his royal power and supremacy over his rivals.
- The rite was performed by a king (or a Brahmana) seeking to attain a higher status and enhanced strength. Its key political symbolism lay in the staged chariot race, in which the king emerged victorious in a contest against kinsmen or rivals, dramatising and reaffirming his dominance. As part of the ceremony, the king also ascended a sacrificial post (yupa), symbolically rising above his people, and underwent rituals affirming his pre-eminence. The whole performance was intended to rejuvenate the king's vitality and consolidate his authority.

Q 52. Ans. C

Explanation:

- The economic condition during Harsha's reign (606–647 CE) is characterised by historians as a period of declining trade and commerce, the decay of urban centres, and the rise of a self-sufficient, localised village economy.
- The decline of trade and commerce is evident from the marked reduction in the number of coins in circulation, the weakening of merchant guilds, and the shrinking of trade centres during and after Harsha's reign. With long-distance and inter-regional trade contracting, the demand for marketable agricultural surplus fell, so villagers began producing mainly for their own local needs. This gave rise to self-sufficient village economies increasingly dependent on agriculture.
- A second major driver was the spread of land grants (to Brahmanas and to officials, who were paid in land rather than cash). The Madhuban and Banskhera copper-plate inscriptions record such grants. Paying officials and religious beneficiaries

through revenue assignments over villages, rather than through coined money, accelerated the decline of a money economy and the localisation of the rural economy, the process scholars term "Indian feudalism," with the kingdom of Harsha often cited as a typical example.

- This is corroborated by the account of the Chinese pilgrim Xuanzang (Hiuen Tsang), who spent several years at Harsha's court. He noted light taxation (the land tax being about one-sixth of the produce) but also recorded that the roads were less safe than in the Gupta period (he himself was robbed), pointing to weakened commercial conditions compared with the earlier classical age.

Q 53. Ans. C

Explanation:

- **Statement 1 is correct:** The first major split (schism) in the Buddhist Sangha is traditionally associated with the Second Buddhist Council held at Vaishali, about a century after the Buddha's death. The dispute over the relaxation of monastic rules (the "ten points" raised by the Vajji/Vriji monks of Vaishali, including accepting gold and silver and eating after noon) led to the division of the Sangha into two schools: the conservative Sthaviravada (Theravada, "the Elders") and the more liberal Mahasanghika (the "Great Assembly"). This split is the standard answer for the first schism in the order.
- **Statement 2 is correct:** The Mahasanghikas developed a more elevated conception of the Buddha, holding that the Buddha was a supramundane, transcendent (lokottara) being and that his earthly existence as Gautama was only an apparition, with a plurality of Buddhas existing throughout the universe. This exalted, quasi-divine view of the Buddha is recognised as a precursor to and a major influence on the later Mahayana tradition.

Q 54. Ans. C

Explanation:

- Nizam-ul-Mulk (Asaf Jah I), one of the most capable nobles of the Mughal Empire, served as the Wazir under Emperor Muhammad Shah. Disillusioned by court intrigues, factional politics, and the declining authority of the central administration, he left the Mughal Court in October 1724 and returned to the Deccan.
  - ✓ His departure is famously described by historians as "symbolic of the flight of loyalty and virtue from the empire."
  - ✓ In the same year, after defeating Mubariz Khan in the Battle of Shakar Kheda (1724), he established the Asaf Jahi dynasty in Hyderabad, while continuing to acknowledge the nominal suzerainty of the Mughal emperor.
  - ✓ The event highlighted the weakening of Mughal central authority and the growing autonomy of provincial governors.

Q 55. Ans. D

Explanation:

- The nineteenth-century social and religious reformers spread their ideas primarily through modern channels of mass communication and popular literature aimed at influencing public opinion, not through the official machinery of princely states. The reformers were largely middle-class, Western-educated intellectuals working through voluntary associations (sabhas and samajes), and royal proclamations were an instrument of state authority, not a reform-movement method.

Q 56. Ans. C

Explanation:

- The Anti-Partition Movement (1905–1911) arose in response to the partition of Bengal by Lord Curzon in 1905. The movement initially remained within the moderate phase, relying on constitutional methods such as petitions, public meetings, resolutions, and peaceful protests.
- Surendranath Banerjea emerged as one of the foremost leaders of this initial moderate phase. He organised meetings, mobilised public opinion, and advocated the annulment of the partition through lawful and constitutional means.
- As the movement progressed, it acquired a more militant character with the emergence of leaders such as Bipin Chandra Pal and Aurobindo Ghosh, who promoted boycott, swadeshi, and passive resistance.

Q 57. Ans. A

Explanation:

- London to publicise Indian grievances and influence British public opinion and Parliament. To further this objective, the Committee started a journal titled India in 1890.
- The journal served as the official organ of the British Committee of the Indian National Congress.
- It highlighted issues such as constitutional reforms,

administrative grievances, economic exploitation, and the demand for greater Indian representation in governance.

- Prominent nationalist leaders such as Dadabhai Naoroji used the British platform to advocate India's cause.

Q 58. Ans. C

Explanation:

- Lord William Bentinck, the Governor-General of India (1828–1835), introduced several administrative and judicial reforms aimed at improving efficiency and reducing expenditure.
- In 1831, he abolished the Provincial Courts of Appeal and Circuit, which had originally been established under Lord Cornwallis.
- Their functions were transferred to Commissioners of Revenue and Circuit, while greater judicial responsibilities were vested in district-level institutions.
- Bentinck also expanded the role of Indian subordinate judges and sought to simplify the judicial system to make it more economical and effective.

Q 59. Ans. A

Explanation:

- The Home Rule Movement, launched in 1916 by Bal Gangadhar Tilak (Indian Home Rule League, April 1916) and Annie Besant (All India Home Rule League, September 1916), emerged in the context of the First World War (1914–1918) and the discontent it generated.
- **The war affected India deeply:** it brought economic hardship through rising prices, heavy taxation, forced recruitment, and the burden of wartime contributions, generating widespread popular discontent that the Home Rule leaders could tap into. At the same time, Indian leaders expected that India's substantial contribution to the British war effort (in men and money) would be rewarded with political concessions and self-government, and the Home Rule agitation was designed to press this demand for self-rule (on the lines of the self-governing dominions). The lull in active Congress agitation during the early war years, and the readmission of the Extremists (Tilak) into the Congress, also created the political space for the movement.

Q 60. Ans. B

Explanation:

- The Montagu-Chelmsford Report was published in July 1918 and became the basis for the Government of India Act, 1919, which introduced dyarchy in the provinces.
- The Lucknow Pact of 1916 united the Congress and the Muslim League on a common set of constitutional demands. This unity, along with India's contribution to the First World War and the pressure of the Home Rule Movement, forced the British to respond. Secretary of State Edwin Montagu made the August Declaration of 1917, promising the gradual development of self-governing institutions. He then toured India with Viceroy Lord Chelmsford, and their report was published in July 1918.

Q 61. Ans. A

Explanation:

- **Statement 1 is correct:** The anti-Rowlatt agitation of 1919 was the first time Gandhi led an all-India mass movement. He called for a nationwide hartal and satyagraha against the Rowlatt Act, which allowed detention without trial, and this marked the beginning of his militant mass-based phase of the national movement.
- **Statement 2 is incorrect:** Non-cooperation was a non-violent method, not a violent one. Gandhi's entire approach was based on satyagraha and peaceful resistance. Calling it a "violent defensive measure" is wrong on both counts, since it was neither violent nor designed as armed defence against the police.

Q 62. Ans. B

Explanation:

- Al Hilal and Comrade were both nationalist Muslim journals suppressed by the colonial government during the First World War.
- Al Hilal was an Urdu weekly founded by Maulana Abul Kalam Azad in 1912. It sharply criticised the British Raj and urged Indian Muslims to join the freedom movement. It was shut down under the Press Act in 1914.
- Comrade was an English weekly published and edited by Maulana Mohammad Ali Jauhar from 1911. It grew increasingly anti-British, especially after Turkey entered the war, and was banned by the colonial authorities in September 1914.

Q 63. Ans. A

Explanation:

- Emperor Jahangir (1605–1627) is particularly renowned for his keen interest in nature and patronage of painting.

- He possessed an extraordinary eye for detail and recorded his observations on animals, birds, plants, and natural phenomena in his memoirs, the Tuzuk-i-Jahangiri (Jahangirnama).
- Jahangir encouraged naturalistic and realistic painting, leading Mughal art to attain a high degree of refinement during his reign.

Q 64. Ans. D

Explanation:

- **Statement 1 is correct:** At the Nagpur Session of 1920, the Provincial Congress Committees were reorganised on a linguistic basis, so that the Congress structure matched the language regions of the country rather than the British administrative provinces. This made the organisation more accessible to the common people.
- **Statement 2 is correct:** The annual Congress membership fee was reduced to four annas, deliberately kept low so that the poor and ordinary people could join the Congress and the movement could become a genuine mass organisation.

Q 65. Ans. C

Explanation:

- Chhatrapati Shivaji Maharaj is most closely associated with the use of guerrilla warfare, popularly known as Ganimi Kava.
- He relied on swift movements, surprise attacks, ambushes, and hit-and-run tactics rather than engaging in prolonged conventional battles.
- The rugged terrain of the Western Ghats, combined with a network of strategically located forts, enabled the Marathas to exploit their knowledge of local geography.

Q 66. Ans. D

Explanation:

- Rajendra Prasad was among the leaders who accompanied and assisted Mahatma Gandhi during his Champaran inquiry in 1917. He was part of the team of lawyers, along with Brajkishore Prasad, Anugrah Narayan Sinha, J.B. Kripalani, Mazharul Haq, and Ramnavami Prasad, who helped record the statements of thousands of peasants and document the exploitation under the tinkathia system.

Q 67. Ans. B

Explanation:

- Mahatma Gandhi founded the Satyagraha Sabha in Bombay in February 1919 specifically to organize a nationwide agitation against the draconian Rowlatt Act.
- Its members took a formal pledge to disobey the Act (and any other unjust laws) and court arrest through Satyagraha (non-violent civil disobedience). This organization was crucial because it allowed Gandhi to bypass the more cautious, moderate leadership of the Indian National Congress at the time.

Q 68. Ans. C

Explanation:

- **Statement 1 is correct:** Jamia Millia Islamia was founded at Aligarh on 29 October 1920, born out of the Khilafat and Non-Cooperation movements. It was established by nationalist teachers and students (led by figures like Maulana Mahmud Hasan and Hakim Ajmal Khan) who left Aligarh Muslim University in response to Gandhi's call to boycott colonial educational institutions. It was later shifted to Delhi (Karol Bagh) in 1925, and eventually to Okhla.
- **Statement 2 is correct:** A major pillar of the Non-Cooperation Movement was the boycott of British schools and the establishment of "National" schools. Eminent figures stepped up as national teachers: Dr. Zakir Husain went on to become a key pillar of Jamia Millia Islamia (and later President of India), while Acharya Narendra Dev became the Vice-Chancellor of Kashi Vidyapeeth, another premier national institution established in Varanasi in 1921.

Q 69. Ans. D

Explanation:

- Motilal Nehru, a leading figure of the Swaraj Party, was accused by the Responsivist faction of being anti-Hindu, of favouring cow slaughter, and of eating beef. These attacks were part of the communal campaign against the Swarajists in the mid-1920s.
- The Responsivists, who included Madan Mohan Malaviya, Lala Lajpat Rai, and N.C. Kelkar, favoured cooperation with the government to safeguard so-called Hindu interests, and they targeted Motilal Nehru because he opposed this line and did not favour their communal approach to council politics.

Q 70. Ans. D

Explanation:

- **Statement 1 is correct:** The No-Changers viewed constructive work, such as the promotion of khadi, national education, Hindu-Muslim unity, and the removal of untouchability, as a way to prepare the masses and the organisation for the eventual revival of mass civil disobedience. They saw it as a necessary foundation for the next phase of the struggle.

- **Statement 2 is correct:** The No-Changers opposed council entry. They argued that participation in legislative politics would lead to the neglect of constructive work among the masses, the loss of revolutionary zeal, and political corruption and factionalism within the nationalist movement.

Q 71. Ans. D

Explanation:

- **Statement 1 is incorrect:** Under Dyarchy, the Government of India Act of 1919 divided subjects into Reserved and Transferred categories. It did not transfer all spending departments along with absolute financial control. The transferred subjects were handed to ministers, but financial control remained limited and constrained, since the finance department stayed under the Reserved half (with the Governor and his Executive Council), so the ministers in charge of transferred subjects lacked independent control over funds.
- **Statement 2 is incorrect:** The provincial Governor did not have complete independent control over provincial finances. The Governor's powers were limited by the dyarchical structure itself, by the provincial legislative council (which voted on the budget and demands for grants), and by the overall control of the Government of India and the Secretary of State. The provinces did not have full financial autonomy under the 1919 Act.

Q 72. Ans. C

Explanation:

- The Paris Peace Conference of 1919, held after the First World War, is where the victorious Allied powers betrayed their wartime promises of national self-determination for the colonies.
- Although US President Woodrow Wilson's Fourteen Points had raised hopes of self-determination, in practice the principle was applied mainly to parts of Europe, while the colonies of Asia and Africa were denied it.
- The German and Ottoman colonies were redistributed among the Allied powers under the mandate system of the League of Nations, which was colonial control in a new form.

Q 73. Ans. B

Explanation:

- The first All-Bengal Conference of Students, held in Calcutta in August 1928, was presided over by Jawaharlal Nehru. This conference led to the formation of the All Bengal Students' Association. Subhas Chandra Bose was present as a guest of honour, but Nehru was the one who chaired it and delivered the inaugural address.

Q 74. Ans. C

Explanation:

- **Statement 1 is incorrect:** The central bicameral legislature (Council of State and Legislative Assembly) set up by the 1919 Act did not exercise complete constitutional control over the Governor-General. The Governor-General retained overriding powers, the power to certify bills the legislature rejected, to veto legislation, and to issue ordinances, and he was not responsible to the legislature. So Statement 1 is clearly wrong.
- **Statement 2 is incorrect:** The central government did not maintain "completely unrestricted" control over all provincial governments. The 1919 Act, for the first time, separated central and provincial subjects and devolved a measure of autonomy to the provinces (this was the beginning of provincial devolution, with dyarchy in the provinces and the transferred subjects placed under ministers responsible to the provincial legislature). The word "completely unrestricted" makes Statement 2 false, because the whole point of the 1919 Act was a partial loosening of central control over the provinces.

Q 75. Ans. C

Explanation:

- In March 1929, the government arrested 31 trade union and communist leaders, including three Englishmen, in the Meerut Conspiracy Case. They were charged under Section 121-A of the Indian Penal Code with conspiracy against British rule. The accused were taken to Meerut for trial (chosen to avoid a jury), and the trial lasted nearly four years, ending in 1933.
- The three Englishmen were Philip Spratt, Benjamin Bradley, and Lester Hutchinson, and prominent Indian leaders included Muzaffar Ahmed, S.A. Dange, and S.S. Joglekar.

Q 76. Ans. A

Explanation:

- **Statement 1 is correct:** The Lakshadweep archipelago consists of exactly 36 coral islands in the Arabian Sea, out of which only 10 are inhabited (Agatti, Amini, Andrott, Bitra, Chetlat, Kadmat, Kalpeni, Kavaratti, Kiltan, and Minicoy).
- **Statement 2 is incorrect:** The Lakshadweep Islands are separated from the Maldive Islands by the Nine Degree Channel. The main Lakshadweep archipelago is separated from Minicoy Island by the Nine Degree Channel, not the Eight Degree Channel. In Lakshadweep, coconut is the only major crop grown. The sea around the island is rich in marine life.

Q 77. Ans. C

Explanation:

- **The Central Himalaya** stretches from the river Kali to the river Tista for about 800 km, occupying an area of about 116,800 sq km. A major part of it lies in Nepal, except the extreme eastern part called Sikkim Himalaya and in the Darjeeling District of West Bengal. All three ranges of the Himalaya are represented here. The highest peaks of the world, like Mt. Everest (8850 m), Kanchenjunga (8598 m), Makalu (8481 m), Dhaulagiri (8172 m), Annapurna (8078 m), Manaslu (8154 m) and Gosainath (8014 m) are situated in this part of the Himalaya.

Q 78. Ans. C

Explanation:

- **Assertion (A) is true:** On 22nd December (Winter Solstice), the Northern Hemisphere experiences its shortest day and longest night of the year because it is tilted away from the Sun. Consequently, a smaller portion of the Northern Hemisphere receives sunlight.
- **Reason (R) is false:** On this date, the Sun's vertical rays fall on the Tropic of Capricorn (23½° S), not on the Tropic of Cancer (23½° N). As the Southern Hemisphere is tilted towards the Sun, it receives greater illumination and experiences its longest day.

Q 79. Ans. D

Explanation:

- **Statement 1 is correct:** The mantle is by far the most voluminous layer of the Earth. While the crust makes up less than 1% of the Earth's volume and the core accounts for about 15%, the mantle occupies roughly 84% of the Earth's total volume and contains approximately 67% of its total mass.
- **Statement 2 is correct:** In terms of chemical composition, the mantle is predominantly composed of solid silicate rocks that are highly enriched in Silica (Si) and Magnesium (Mg), along with Iron. In classical geological classifications (often tested in competitive examinations), the oceanic crust and the underlying mantle are grouped under the term SIMA (Silica + Magnesium), distinguishing it from the lighter continental crust known as SIAL (Silica + Aluminium).
- **Statement 3 is correct:** Density within the Earth increases strictly with depth due to extreme pressure and heavier mineral compositions.
  - ✓ The continental crust has an average density of about 2.7 g/cm^3.
  - ✓ Just below the crust, at the Mohorovi→ (Moho) discontinuity, the mantle begins with a density of about 3.3 g/cm^3. As you travel deeper through the upper and lower mantle towards the core-mantle boundary (the Gutenberg discontinuity), the density steadily increases to roughly 5.4 to 5.5 g/cm^3.

Q 80. Ans. C

Explanation:

- The Pamir Plateau is popularly known as the "Roof of the World" because of its extremely high elevation and its position as a meeting point of several major mountain ranges, including the Himalayas, Karakoram, Hindu Kush, Kunlun Mountains, and Tien Shan.

Q 81. Ans. A

Explanation:

- **Statement 1 is correct:** In geomorphology, potholes are more or less circular, cylindrical depressions carved into the solid rocky beds of hill streams and rivers. They are formed when the turbulent flow of water creates eddies (swirling currents) that catch pebbles, gravel, and small boulders, using them as drilling tools to grind into the bedrock.
- **Statement 2 is incorrect:** The primary mechanism behind the formation and deepening of potholes is vertical corrasion (downward drilling or abrasion), not lateral corrasion. While lateral corrosion does occur and widens the hole over time, the driving force that initially drills the depression into the rock is vertical.

- **Statement 3 is correct:** Once a shallow depression is formed, more rock fragments get trapped inside. The continued swirling and rotation of these pebbles and boulders by the flowing water acts like a drill, progressively enlarging the potholes in both depth and width. Over time, multiple adjacent potholes can coalesce (merge) to form much larger features, such as plunge pools at the base of waterfalls.

Q 82. Ans. B

Explanation:

- The term "Cretaceous" is derived from the Latin word creta, meaning "chalk." The name was coined because extensive deposits of chalk, a soft white limestone composed mainly of microscopic marine organisms, were formed during this geological period.
- The Cretaceous Period was the last period of the Mesozoic Era, extending from about 145 million to 66 million years ago.
- It is noted for widespread chalk formations, such as the famous White Cliffs of Dover in England.
- The period ended with the Cretaceous–Paleogene (K–Pg) mass extinction event, which led to the extinction of non-avian dinosaurs.

Q 83. Ans. B

Explanation:

- **Statement 1 is correct:** Resequent rivers are tributary streams that develop later but flow in the same direction as the original consequent rivers, generally following the regional slope. In contrast, obsequent rivers flow in the opposite direction to the consequent streams, often developing along weaker rock strata.
- **Statement 2 is incorrect:** Resequent rivers are not older features formed early in the erosion cycle. They develop during the later stages of landscape evolution as erosion progresses and new drainage patterns emerge. Obsequent rivers also arise subsequently; therefore, the distinction between them is based on the direction of flow, not their relative age.

Q 84. Ans. D

Explanation:

- The shifting of atmospheric circulation over the southern Pacific region is known as Southern Oscillation. It is a precursor to the occurrence of an El-Nino event. In an El-Nino year, a major change occurs across the entire stretch of the equatorial zone as far west as south-eastern Asia. Normally, low pressure prevails over northern Australia, New Guinea, and Indonesia, where the largest and warmest body of ocean water can be found. This low pressure is associated with a pool of warm water and a deep thermocline

Q 85. Ans. D

Explanation:

- **Statement 1 is incorrect:** Rivers flowing outward from a central highland exhibit a radial drainage pattern, not a parallel drainage pattern. In parallel drainage, rivers flow almost parallel to one another over uniformly sloping surfaces.
- **Statement 2 is incorrect:** The Chambal, Banas and Kali Sindh belong to the Yamuna–Ganga drainage system and are not examples of parallel drainage. In India, parallel drainage is typically represented by the short west-flowing rivers originating in the Western Ghats, such as the Sharavati, Kalinadi, Zuari and Periyar.

Q 86. Ans. A

Explanation:

- A Meso-scale Convective Complex (MCC) is a highly organised, large-scale cluster of thunderstorms. In meteorology, it is a specific, well-defined subset of Mesoscale Convective Systems (MCS).
  - ✓ **Formation:** It occurs when multiple individual, single-cell or multi-cell thunderstorms merge into a single, massive, and highly organised convective weather system.
  - ✓ **Characteristics:** To be officially classified as an MCC by meteorological organisations (like the WMO), the system must produce a continuous, nearly circular cloud shield of a specific massive size (usually over 100,000 square kilometres) and must persist for an extended period, typically greater than 6 to 12 hours.

Q 87. Ans. A

Explanation:

- **Statement 1 is correct:** Bioaccumulation refers to the build-up of pollutants within an organism through direct absorption from water, air, soil, or food. Therefore, even organisms at the lowest trophic levels can exhibit bioaccumulation.
- **Statement 2 is incorrect:** A progressive increase in pollutant concentration across trophic levels is the defining feature of biomagnification, not

bioaccumulation. Bioaccumulation focuses on the net accumulation within an individual organism over time.

Q 88. Ans. B

Explanation:

- Oxford is called the "City of Dreaming Spires." The phrase was coined by the English poet Matthew Arnold in his poem "Thyrsis" (1865), referring to the beautiful skyline formed by the spires and towers of the University of Oxford's historic college buildings.

Q 89. Ans. C

Explanation:

- Baratang Island is separated from Middle Andaman Island by Homfray's Strait, a shallow, narrow channel only a few hundred metres wide. To its south, Baratang is separated from South Andaman Island by the Andaman Middle Strait.

Q 90. Ans. B

Explanation:

- Dhosi Hill, located on the Haryana-Rajasthan border at the north-western end of the Aravalli range (Mahendragarh district, Haryana, and Jhunjhunu district, Rajasthan), is an extinct volcano. It belongs to the Precambrian Malani igneous suite, with volcanic activity dating back to about 732 million years ago. The hill still shows a distinct crater and remnants of ancient lava flows, but it has had no recorded activity in historical times, which is why it is classified as extinct.

Q 91. Ans. A

Explanation:

- **Statement 1 is correct:** Article 21, which protects the right to life and personal liberty, is available to all persons, including citizens and foreigners, and not merely to citizens; it provides that no person shall be deprived of life or personal liberty except according to procedure established by law.
- **Statement 2 is correct:** In the Maneka Gandhi case of 1978 the Supreme Court significantly expanded the scope of Article 21 by holding that the procedure established by law must be fair, just, and reasonable, thereby introducing the substance of due process and linking Articles 14, 19, and 21; the right to life has since been read to include the right to live with dignity, livelihood, privacy, a clean environment, health, education, and shelter.
- **Statement 3 is incorrect:** Article 21 cannot be suspended even during a national emergency, a safeguard made absolute by the 44th Amendment Act of 1978, so that the right to life and personal liberty and the protection under Article 20 remain enforceable at all times. Article 21A, added by the 86th Amendment, provides the right to free and compulsory education to children between six and fourteen years.

Q 92. Ans. C

Explanation:

- **Assertion is true:** The framers of the Constitution deliberately preferred the parliamentary system over the presidential system, both at the Centre and in the states, mainly because of their familiarity with the parliamentary model under British rule and their preference for a system in which the executive is responsible to the legislature.
- **Reason is false:** The parliamentary system does not provide greater stability; on the contrary, instability is one of its principal demerits, since the government can fall whenever it loses the confidence of the lower house. It is the presidential system that offers greater stability through a fixed tenure of the executive. The framers chose the parliamentary model for responsibility rather than stability.
- The merits of the parliamentary system include harmony between the legislature and the executive, responsible government, the prevention of despotism, and wide representation, while its demerits include unstable government, discontinuity of policies, a violation of the strict separation of powers, and the danger of cabinet dictatorship. India adopted it also because it provides for the representation of diverse groups and avoids conflicts between the legislature and the executive that are common in the presidential system.

Q 93. Ans. A

Explanation:

- **Statement 1 is correct:** Under Article 61 the President can be removed from office by a process of impeachment for the violation of the Constitution, though the Constitution does not define the expression violation of the Constitution.
- **Statement 2 is correct:** The impeachment charge can be preferred by either House of Parliament; such a charge must be signed by at least one fourth

of the total members of the House, a fourteen day notice must be given, and the resolution must then be passed by a majority of two thirds of the total membership of that House, after which the other House investigates the charge.

- **Statement 3 is incorrect:** The nominated members of either House can participate in the impeachment of the President, even though they do not take part in the election of the President. By contrast, the elected members of the state legislative assemblies, who participate in the election, have no role in the impeachment. Under Article 59 the President must not hold any other office of profit and is entitled to emoluments and the use of an official residence.

Q 94. Ans. C

Explanation:

- **Statement 1 is correct:** The present sanctioned strength of the Supreme Court is the Chief Justice of India and thirty three other judges, making a total of thirty four, the strength having been raised to this figure in 2019; Parliament is empowered to regulate the number of judges. **However, in 2026, the sanctioned strength was increased to 38 judges (Chief Justice + 37 other judges) through the Supreme Court (Number of Judges) Amendment Ordinance, 2026. As of June 2026, the Court is functioning with 37 judges, including the Chief Justice, with one vacancy remaining.**
- **Statement 2 is incorrect:** A judge of the Supreme Court cannot be removed on the advice of the Council of Ministers; under Article 124 a judge can be removed only by an order of the President passed after an address by both Houses of Parliament, supported in each House by a special majority, on the ground of proved misbehaviour or incapacity, the procedure being regulated by the Judges Inquiry Act of 1968.
- **Statement 3 is correct:** The salaries, allowances, and pensions of the judges are charged on the Consolidated Fund of India and are not subject to the vote of Parliament, which is one of the safeguards securing the independence of the judiciary, along with security of tenure, a ban on practice after retirement, the power to punish for contempt, and a prohibition on the discussion of their conduct in Parliament except on a removal motion.

Q 95. Ans. C

Explanation:

- Under Article 169 a Legislative Council can be created in a state that does not have one, or an existing Council can be abolished, by a law of Parliament, and such a law is not deemed to be a constitution amendment under Article 368 and is passed by a simple majority of Parliament.
- Parliament can act only if the legislative assembly of the state concerned first passes a resolution to that effect by a special majority, that is a majority of the total membership of the assembly and a majority of not less than two thirds of the members present and voting, and not by a simple majority.
- The Legislative Council is a continuing chamber that is not subject to dissolution, one third of its members retiring every two years. It is a weaker and subordinate chamber, as its strength must be between one third of the assembly's strength and forty, and its members are partly elected by various constituencies and partly nominated by the Governor for contributions to literature, science, art, the cooperative movement, and social service.

Q 96. Ans. B

Explanation:

- Article 1 describes India, that is Bharat, as a Union of States, and the expression Union was deliberately preferred to Federation to indicate that the Indian federation is not the result of an agreement among the states and that the states have no right to secede from the Union.
- Article 2 deals with the admission or establishment of new states that are not already part of the Union of India, whereas the alteration of the areas, boundaries, or names of existing states is dealt with under Article 3. The territory of India is wider than the Union of States, as it includes the states, the union territories, and any territories that may be acquired.

Q 97. Ans. D

Explanation:

- The scope of judicial review in India is narrower than in the United States, because the American Constitution provides for a broader review through the due process of law clause, whereas the Indian Constitution adopts the expression procedure established by law, which originally permitted a narrower review.
- Indian courts ordinarily follow the doctrine of procedure established by law and not the American doctrine of due process of law; however, after the Maneka Gandhi case the Supreme Court has

read the substance of due process into Article 21 by requiring that any procedure be fair, just, and reasonable.

- Judicial review is the power of the courts to examine the constitutionality of legislative and executive actions and to declare them void if they are unconstitutional. The grounds on which a law may be challenged are that it infringes the fundamental rights, that it is outside the competence of the authority that enacted it, or that it is repugnant to the constitutional provisions. The importance of judicial review lies in upholding the supremacy of the Constitution, maintaining the federal balance, and protecting the fundamental rights.

Q 98. Ans. A

Explanation:

- **Statement 1 is correct:** Public interest litigation can be filed by any public spirited person or social action group on behalf of those who are poor, illiterate, or otherwise disadvantaged and are unable to approach the court themselves, which marks a departure from the rule that only an aggrieved person can seek relief.
- **Statement 2 is correct:** Public interest litigation dilutes the traditional rule of locus standi by permitting persons who are not themselves directly injured to approach the court in the interest of the public, a relaxation introduced by the judiciary to widen access to justice.
- **Statement 3 is incorrect:** Public interest litigation can be filed not only before the Supreme Court under Article 32 but also before a High Court under Article 226, and the courts have even treated letters and telegrams as petitions under what is called epistolary jurisdiction. To prevent its misuse as publicity or private interest litigation, the courts have laid down guidelines requiring a genuine public cause and discouraging frivolous or motivated petitions.

Q 99. Ans. A

Explanation:

- The All India Services, namely the Indian Administrative Service, the Indian Police Service, and the Indian Forest Service, are common to the Centre and the states, their members serving the Union and the states by turns and occupying the higher posts in both.
- Under Article 312 a new All India Service can be created only if the Rajya Sabha first passes a resolution, supported by not less than two thirds of the members present and voting, declaring that it is necessary or expedient in the national interest to do so, after which Parliament can make a law creating the service.
- Members of the All India Services are not recruited and controlled solely by the states; they are recruited and trained by the Centre through the Union Public Service Commission, are allotted to various state cadres, and are subject to the ultimate control of the Centre, even though they serve under the states. Services are classified into All India Services, Central Services, and State Services, and the doctrine of pleasure under Article 310 is qualified by the safeguards in Article 311.

Q 100. Ans. B

Explanation:

- In a pure, classical federation (like the United States), the upper house of the legislature grants absolute equality to all states regardless of their size or population (e.g., every US state gets exactly 2 Senators). This acts as a strict safeguard for federalism, ensuring small states cannot be bullied by large ones.
- In India, however, seats in the Rajya Sabha are allocated based on population (Schedule IV of the Constitution). As a result, highly populated states (like Uttar Pradesh with 31 seats) heavily dominate smaller states (like Tripura with 1 seat). This structural inequality leans power heavily toward the demographic center, which constitutional scholars classify as a distinct unitary (non-federal) bias in the Indian Constitution.

Q 101. Ans. B

Explanation:

- **The Increasing Role of Media:** In contemporary Indian electoral politics, electronic media, 24/7 news channels, and especially targeted social media platforms (WhatsApp, X/Twitter, Facebook) have emerged as the most dominant forces in molding public opinion, setting political narratives, and mobilizing voter bases.

Q 102. Ans. A

Explanation:

- **Statement 1 is correct:** Non-alignment, the cornerstone of India's foreign policy during the Cold War, meant keeping away from the rival military blocs led by the United States and the

Soviet Union, and retaining the freedom to judge each international issue on its merits.

- **Statement 2 is correct:** India's nuclear doctrine, adopted in 2003, is based on a posture of no first use, along with credible minimum deterrence, a commitment not to use nuclear weapons against non nuclear weapon states, and civilian political control over the nuclear arsenal.
- **Statement 3 is incorrect:** The Look West policy concerns India's engagement with the countries of West Asia and the Gulf, not Central Asia; engagement with the Central Asian republics is pursued through the Connect Central Asia policy. The broad objectives of India's foreign policy are the preservation of national security, the promotion of economic development, and the maintenance of world peace, guided by principles such as anti colonialism, Panchsheel, and the promotion of international peace and security under Article 51.

Q 103. Ans. A

Explanation:

- A bill contemplating the formation of new states or the alteration of existing ones can be introduced only on the prior recommendation of the President, who must first refer it to the legislature of the affected state for its views, which are not binding. The 100th Amendment Act of 2015 gave effect to the exchange of certain enclaves with Bangladesh under the land boundary agreement.
- **97th Amendment (2011):** Gave constitutional status and protection to Cooperative Societies.
- **99th Amendment (2014):** Established the National Judicial Appointments Commission (NJAC), later struck down by the Supreme Court.
- **101st Amendment (2016):** Introduced the Goods and Services Tax (GST).

Q 104. Ans. B

Explanation:

- **Statement 1 is correct:** Article 39A, which directs the State to promote equal justice and to provide free legal aid to the poor, was one of the new directive principles added by the 42nd Amendment Act of 1976.
- **Statement 2 is correct:** Article 43A, which directs the State to take steps to secure the participation of workers in the management of industries, was also added by the 42nd Amendment Act of 1976.
- **Statement 3 is correct:** Article 48A, which directs the State to protect and improve the environment and to safeguard forests and wildlife, was likewise added by the 42nd Amendment Act of 1976.
- **Statement 4 is incorrect:** Securing a uniform civil code under Article 44 is an original directive principle and was not added by the 42nd Amendment. The 44th Amendment later added Article 38(2) on the reduction of inequalities, the 86th Amendment substituted Article 45 to deal with early childhood care, and the 97th Amendment added Article 43B on the promotion of cooperative societies.

Q 105. Ans. B

Explanation:

- **Statement 1 is correct:** Judicial review has been held to form part of the basic structure, notably in the Minerva Mills case, since it is essential for upholding the supremacy of the Constitution and protecting the fundamental rights.
- **Statement 2 is correct:** The federal character of the Constitution has been recognised as part of the basic structure, so that the distribution of powers between the Centre and the states cannot be destroyed by amendment.
- **Statement 3 is correct:** Free and fair elections have been held to form part of the basic structure, as they are fundamental to the democratic and republican character of the polity.
- **Statement 4 is incorrect:** The right to property is not part of the basic structure; it was deleted as a fundamental right by the 44th Amendment and made an ordinary legal right under Article 300A. Other elements of the basic structure include the supremacy of the Constitution, the rule of law, the separation of powers, the secular character of the State, the independence of the judiciary, and the balance between fundamental rights and directive principles.

Q 106. Ans. B

Explanation:

- The ND-GAIN (Notre Dame Global Adaptation Initiative) Country Index measures a country's vulnerability to climate change and its readiness to improve resilience and adapt.
- The vulnerability dimension assesses a country's exposure, sensitivity, and capacity to adapt across six life-supporting sectors (food, water, health, ecosystem services, human habitat, and infrastructure), while the readiness dimension measures its economic, governance, and social ability to convert investments into adaptation actions.

- Combining both gives a single score that captures both the risk a country faces and its preparedness to respond.

Q 107. Ans. C

Explanation:

- According to UNCTAD's Trade and Development Report 2025, India's trade diversity score (the diversity index of trade partnerships) was higher than that of all Global North economies. India ranked third among Global South economies on this index, after China and the UAE, with a score of 3.2. This reflects the resilience of India's trade ecosystem in the face of tariff uncertainties, supported by an expanding network of free trade agreements and a diversified base of trading partners.

Q 108. Ans. C

Explanation:

- **Statement 1 is correct:** The Merchandise Balance of Trade (also called the visible balance of trade) is the difference between a country's exports of goods and its imports of goods. If exports of goods exceed imports of goods, there is a trade surplus; if imports exceed exports, there is a trade deficit.
- **Statement 2 is correct:** The Services Balance of Trade (also called the invisible balance of trade) is the difference between a country's exports of services and its imports of services. Services include items such as software, tourism, transport, insurance, and financial services.

Q 109. Ans. C

Explanation:

- **Personal Income is the part of National Income that is actually received by households. The formula is:**
  - ✓ Personal Income (PI) = National Income – Undistributed profits – Net interest payments made by households – Corporate tax + Transfer payments to the households from the government and firms.
- Transfer payments to households (such as pensions, scholarships, unemployment benefits, and gifts) are added to National Income because households receive this income even though it is not a payment for any current productive service. So option 3 is the item that is added.

Q 110. Ans. A

Explanation:

- **Statement 1 is correct:** In national income accounting, the aggregate expenditure on final goods and services is equal to the aggregate value of final output produced in the economy. This equality arises because every unit of final output produced is ultimately purchased by some economic agent. Thus, the expenditure method and the output (value-added) method of measuring national income yield the same result.
- **Statement 2 is incorrect:** Aggregate income consists of the factor incomes earned from the production process, such as:
  - ✓ Wages and salaries (payment to labour),
  - ✓ Rent (payment for land and other property),
  - ✓ Interest (payment for capital), and
  - ✓ Profits (reward to entrepreneurs).
- However, transfer payments are not included in aggregate income because they are received without any corresponding contribution to current production. Examples include pensions, unemployment benefits, scholarships, and old-age assistance. Since these payments do not arise from the production of goods and services, they are excluded from national income.

Q 111. Ans. C

Explanation:

- Electronics emerged as India's fastest-growing export category, recording a year-on-year growth of approximately 33.6% (2024 to 2025), driven primarily by mobile phone and component exports under the Production Linked Incentive (PLI) scheme. This marks a structural shift in India's export basket from traditional commodities like petroleum and gems toward high-value electronics manufacturing.

Q 112. Ans. B

Explanation:

- The Kisan Rin Portal (KRP) was launched by the Indian government to bring transparency and efficiency to the agricultural credit system, specifically focusing on the Kisan Credit Card (KCC) scheme.
- Its primary objective is to serve as a centralized digital platform where banks, government departments, and financial institutions can track farmer loan data in real time. It digitizes the entire process of managing KCC loans, monitoring how much credit is disbursed, tracking repayment records, and processing "interest subvention" (the government subsidy that lowers the interest rate for farmers who repay their loans on time).

Q 113. Ans. A

Explanation:

- According to the PLFS Annual Report 2025 (January–December 2025), the Labour Force Participation Rate (LFPR) in usual status for persons aged 15 years and above was 79.1% for males and 40.0% for females, giving a male-female gap of 39.1 percentage points (79.1 – 40.0).
- On the reason for non-participation, the report states that among women, 44.4% cited child care and personal commitments in home-making as the main reason for not being in the labour force, the most cited reason. (Among men, by contrast, 69.8% reported wanting to continue their studies as the main reason.)

Q 114. Ans. B

Explanation:

- **Statement 1 is correct:** The firm supply curve represents how a firm responds to different market prices while trying to maximise profit. At each possible price level, the firm chooses the quantity of output it is willing to supply. Therefore, it is correctly described as the "best response of sellers to different market prices."
- **Statement 2 is incorrect:** This statement wrongly states the axis convention used in economics diagrams.
- **In standard economic graphs (including supply and demand curves):**
  - ✓ Quantity is measured on the horizontal (X) axis
  - ✓ Price is measured on the vertical (Y) axis

Q 115. Ans. C

Explanation:

- **Statement 1 is correct:** This is the fundamental definition of the law in economics. If you keep at least one production input fixed (like the size of a factory or a piece of land) and continually add more of a variable input (like workers), the extra output produced by each new worker will eventually start to decrease.
- **Statement 2 is correct:** Because the law dictates that marginal product will eventually fall, it serves as a crucial metric for business managers. It helps them determine the optimal level of production and hiring. Once the marginal product of an additional unit of input drops so low that its contribution is worth less than its cost, adding more input becomes economically inefficient and unprofitable.

Q 116. Ans. D

Explanation:

- **In public health and epidemiology, health indicators and determinants are generally categorized based on how directly they impact a person's physical health:**
- **Distal (Upstream) Indicators:** These are the foundational, structural, or socio-economic factors that influence health indirectly or "from a distance." They shape a person's living conditions, lifestyle choices, and access to resources over time. Examples include income, occupational status, housing quality, and adult literacy rate. Education and literacy are classic distal indicators because they dictate an individual's ability to understand health information, secure safe employment, and navigate healthcare systems.
- **Proximal (Downstream) Indicators:** These are direct, immediate biological factors or specific health outcomes. They represent the actual manifestation of health or disease in a population.

Q 117. Ans. C

Explanation:

- According to UNCTAD's Trade and Development Report 2025, India ranked third among Global South economies in trade partner diversification (the diversity index of trade partnerships), after China and the UAE. India's index score of 3.2 was higher than that of all Global North economies, reflecting the resilience of its trade ecosystem amid tariff uncertainties and an expanding network of free trade agreements.

Q 118. Ans. B

Explanation:

- The cement industry is a classic example of a "weight-losing" industry. The primary raw materials required to manufacture cement, limestone, coal, and gypsum are incredibly bulky and heavy.
- Furthermore, the final product (cement) has a very low value-to-weight ratio, meaning it is heavy and expensive to transport relative to its actual market price.
- **To maintain profitability and minimize exorbitant freight costs, it is an economic necessity to set up integrated cement plants as close as possible to the source of the heaviest raw material:** limestone.
- India is actually highly self-sufficient in limestone, with massive domestic reserves (especially in states

like Rajasthan, Madhya Pradesh, and Andhra Pradesh), so the industry is not dependent on imported inputs or coastal locations.

- The cement industry is highly polluting and strictly requires stringent environmental clearances to operate.
- Because integrated cement plants must be located near raw materials, major cement production in Bihar is almost entirely geographically restricted to Bihar's southwestern edge, specifically the Rohtas and Kaimur districts (such as the plants in Banjari and Dalmianagar). This is the only region in Bihar with significant limestone deposits, as it forms the edge of the Vindhyan rock system.

Q 119. Ans. C

Explanation:

- Under the Production Linked Incentive (PLI) scheme, manufacturers receive incentives primarily on the basis of **incremental sales** of products manufactured in India over a **defined base year**.
- The scheme pays a percentage-based incentive on the additional (incremental) sales generated above the base-year threshold, which is what makes it "production-linked."
- This output-and-sales-based design is meant to reward actual manufacturing performance and scaling up, rather than merely investment or facility creation.
- **Total capital investment:** While making a minimum threshold of cumulative capital investment is usually a strict eligibility requirement to enter the scheme, the actual financial payout is calculated based on the sales generated from that investment, not the investment amount itself.
- **Number of new manufacturing facilities:** The scheme does not incentivise the mere construction of buildings; it rewards the actual output and sale of products from those facilities.
- **Export performance:** While the PLI scheme ultimately aims to make Indian products globally competitive and boost exports, the incentive is paid on total incremental sales of domestically manufactured goods, whether those goods are sold within India or exported abroad.

Q 120. Ans. A

Explanation:

- The problem of double counting arises when the value of intermediate goods and services is counted along with the value of final goods and services while estimating national income. Since the value of intermediate goods is already embodied in the price of the final product, including both leads to an overestimation of national income.
- For example, if the value of wheat, flour, and bread are all added separately while calculating national income, the value of wheat and flour gets counted more than once because they are intermediate goods used in producing bread, the final good.

Q 121. Ans. A

Explanation:

- **Assertion (A) is true: Diara lands** are low-lying alluvial tracts formed along the floodplains of major rivers such as the **Ganga, Gandak, Kosi, Ghaghara, and Son**. These regions are highly dependent on the annual monsoon because river discharge, flooding intensity, erosion, and fresh sediment deposition are directly influenced by monsoon rainfall.
- **Reason (R) is true:** Diara regions lie within **active floodplains**, where rivers frequently change their course, deposit sediments, and reshape the landscape. The annual behavior of rivers—particularly during the monsoon season—determines the extent of flooding, silt deposition, bank erosion, and land formation in these areas.
- **Why is the Reason the Correct Explanation?**
  - ✓ The very reason Diara regions are sensitive to monsoon fluctuations is that they are situated in active floodplains. Since river flow and flood levels depend heavily on monsoon rainfall, any variation in monsoon performance directly affects these regions. Thus, the location of Diara lands in dynamic riverine environments explains their high sensitivity to monsoon conditions.

Q 122. Ans. D

Explanation:

- **Statement 1 is correct:** Takht Sri Patna Sahib stands at the birthplace of **Guru Gobind Singh (1666)**, the tenth Sikh Guru. The original shrine existed earlier, but the present grand structure was largely rebuilt and patronized by **Maharaja Ranjit Singh**, the founder of the Sikh Empire. The shrine was visited by Guru Nanak Dev, Guru Tegh Bahadur, and Guru Gobind Singh.
- **Statement 2 is correct:** Takht Sri Patna Sahib is one of the **five Takhts (temporal seats of authority)** in Sikhism. It holds exceptional importance because it marks the birthplace of Guru Gobind Singh.

Among the five Takhts, the **Akal Takht** at Amritsar is considered the highest seat of temporal authority in Sikhism. After the **Akal Takht, Takht Sri Patna Sahib** is widely regarded as the **second most revered Takht** due to its direct association with Guru Gobind Singh and its central role in Sikh religious tradition.

- **Statement 3 is correct:** The architecture of Takht Sri Patna Sahib represents a blend of **Rajput and Mughal architectural traditions**. The shrine is constructed largely from **white marble** and features domes, arches, decorative motifs, and intricate carvings that reflect both architectural influences.

Q 123. Ans. D

Explanation:

The Bihar Government announced on 16 February 2026 its plan to develop several dams and reservoirs as tourism destinations to promote dam tourism, water-based recreation, environmental conservation, and local employment generation. Among the identified sites are Durgawati Reservoir, Kundaghat Reservoir, Morve Reservoir, and Odhani Dam.

- **Pair 1 is correct:** The **Durgawati Reservoir**, also known as **Karamchat Dam**, is situated in the border region of **Kaimur and Rohtas districts**. Built across the Durgawati River, it serves irrigation purposes and is one of the important reservoir projects of south-west Bihar.
- **Pair 2 is correct:** The **Kundaghat Reservoir** is located in **Jamui district**, specifically in the **Sikandra Block**. The reservoir has been created by constructing a dam across the **Bahuara (Bahaur) River** and is an important water resource structure in the district.
- **Pair 3 is correct:** The **Morve Reservoir (Morve Dam)** is situated in the **Chanan Block of Lakhisarai district**. It is used for irrigation, fisheries, and other local livelihood activities and has been included in the state's reservoir tourism development initiative.
- **Pair 4 is correct: Odhani Dam** is located in **Banka district**, near **Kakwara Tola Gangti village** on the **Orhni River**. The site has emerged as a local tourism destination and offers recreational activities such as **boating and jet skiing**.

Q 124. Ans. D

Explanation:

- **Statement 1 is incorrect:** The **Jal-Jeevan-Hariyali Mission (JJHM)** was launched by the **Government of Bihar on 2 October 2019**, under the leadership of **Chief Minister Nitish Kumar**, and not in 2024. The mission was conceived as a flagship environmental and climate-action programme to address challenges such as **climate change, declining groundwater levels, shrinking green cover, and environmental degradation.**
- **Statement 2 is incorrect:** The mission is **not limited to South Bihar or drought-prone districts**. It is being implemented across **all 38 districts of Bihar**.
- **Its statewide objectives include:**
  - ✓ **Water conservation and rejuvenation** of ponds, wells, check dams, and irrigation channels.
  - ✓ **Afforestation and plantation drives** to increase Bihar's green cover.
  - ✓ Promotion of **sustainable and climate-resilient agriculture**.
  - ✓ Encouraging **drip irrigation, organic farming, and alternative cropping systems.**
  - ✓ Promotion of **solar energy and rainwater harvesting**.
  - ✓ Construction of **groundwater recharge structures and soak pits**.
  - ✓ Removal of encroachments from public water bodies.
  - ✓ Convergence with **MGNREGA** to create rural employment while developing environmental assets.

Q 125. Ans. A

Explanation:

- **Statement 1 is correct:** The **Bihar Legislative Assembly celebrated its 105th Foundation Day on 7 February 2026** in Patna. The occasion commemorated the evolution of Bihar's legislative traditions and democratic institutions over more than a century.
- **Statement 2 is correct:** The Foundation Day is observed on **7 February** because on this date in **1921**, the **first legislative session of Bihar** was convened. The 2026 celebration therefore marked **105 years** of legislative functioning in the state.
- **Statement 3 is incorrect:** During the Foundation Day celebrations, **Lok Sabha Speaker Om Birla** formally launched the **National e-Vidhan Application (NeVA)** at the Bihar Legislative Assembly.
- The launch marked Bihar's transition towards a **paperless legislative system**, enabling legislators to access proceedings and documents digitally.
  - ✓ **National e-Vidhan Application (NeVA)** is a **Unicode-compliant digital platform** designed to modernize and enable **paperless functioning of legislatures** across India. It

provides bilingual access (English and regional languages) to legislative documents such as the **List of Questions, List of Business, Bills, Reports, and other Assembly records** through web and mobile platforms. NeVA operates on the principles of **"One Nation–One Application," "Cloud First," "Mobile First," and "Members' FIRST,"** aiming to create a unified, cloud-based, and legislator-centric digital ecosystem for all State Legislatures and Union Territories.

The **Bihar Legislative Assembly (Vidhan Sabha)** is the **lower house of the bicameral legislature of Bihar**. It currently functions under the constitutional framework with **Syed Ata Hasnain as the Governor**, **Prem Kumar (BJP) as the Speaker**, **Samrat Choudhary (BJP) as the Leader of the House (Chief Minister)**, and **Tejashwi Yadav (RJD) as the Leader of the Opposition**. The Assembly consists of **243 directly elected members**, chosen through the **first-past-the-post electoral system**. In accordance with constitutional provisions for political representation, **38 seats are reserved for Scheduled Castes (SCs)** and **2 seats for Scheduled Tribes (STs)**, while the remaining **203 seats are unreserved (General category)**.

Q 126. Ans. C

Explanation:

- Nitya Pandey, a 13-year-old boxer from Pipra village in Aurangabad district, Bihar, earned selection to represent India in the 52–55 kg category at the Asian Boxing U15 & U17 Championships 2026. She secured her place in the national squad by finishing first in her weight category during the national selection trials held at Patiala, thereby earning an international berth for the prestigious continental championship.

Q 127. Ans. C

Explanation:

- **Assertion (A)** is true. The forested districts of **Kaimur, Rohtas, Gaya, Nawada, Jamui, Munger, Banka, and parts of Lakhisarai** in southern Bihar are characterized by **tropical dry deciduous forests**, with **Sal (Shorea robusta)** as an important species in suitable habitats. These forests shed their leaves during the dry season to conserve moisture and are adapted to the climatic conditions of the southern plateau and hill regions of Bihar.
- **Reason (R) is False:** The **Terai belt of northern Bihar**, particularly districts such as **West Champaran, Kishanganj, Araria, Purnea, Supaul, and parts of East Champaran**, receives **higher rainfall** than most districts of southern Bihar. In fact:
  - ✓ **Kishanganj** records the highest annual rainfall in Bihar (often exceeding 2,000 mm).
  - ✓ The Himalayan foothill and Terai regions receive heavy monsoon rainfall.
  - ✓ Southern districts such as Kaimur, Rohtas, Gaya, Jamui, and Banka generally receive **lower to moderate rainfall** and are relatively more drought-prone.

Q 128. Ans. A

Explanation:

- The Confederation of Indian Industry (CII) conferred the prestigious CII Quality Ratna Award 2025 upon Venu Srinivasan during the 33rd CII Excellence Summit held in Kolkata. This award recognizes his decades-long pioneering leadership in implementing Total Quality Management (TQM) and driving manufacturing excellence in India.
- Venu Srinivasan is the Chairman Emeritus of TVS Motor Company (and auto components manufacturer TVS Holdings), not the Essar Group. Under his corporate leadership, TVS Motor became the first two-wheeler company in the world to win the coveted Deming Application Prize in 2002

Q 129. Ans. D

Explanation:

- The Indian Insurance Companies (Foreign Investment) Amendment Rules, 2025, which followed the passage of the Sabka Bima Sabki Raksha (Amendment of Insurance Laws) Act of 2025, introduced a landmark reform by increasing the maximum permissible Foreign Direct Investment (FDI) limit in the Indian insurance sector from 74% to 100%.
- This allows foreign investors to fully own and control Indian insurance companies under the automatic route. The reforms were designed to modernize the sector, attract global capital, facilitate the transfer of advanced technologies and best practices, and increase overall insurance penetration as part of the broader national vision of "Insurance for All by 2047."

Q 130. Ans. C

Explanation:

- The launch of the BlueBird Block-2 satellite (designated as the LVM3-M6 mission) marked the sixth operational flight of ISRO's heavy-lift LVM3 rocket. In addition to this milestone, it was also the third dedicated commercial mission for the LVM3. Weighing approximately 6,100 kg, the BlueBird Block-2 spacecraft became the heaviest commercial payload ever launched from Indian soil into Low Earth Orbit (LEO).

Q 131. Ans. B

Explanation:

- The Kalpetta judicial district in Wayanad, Kerala, recently achieved the historic milestone of becoming India's first fully paperless district court system.
- The completely digital court system was inaugurated virtually by Chief Justice of India, Surya Kant, in January 2026.
- Under this system, all courts in the district judiciary operate entirely in digital mode. Every stage of the judicial process, from the initial e-filing of cases to pre-trial proceedings, recording of evidence, interlocutory applications, and the delivery of final judgments, is conducted electronically without the need for physical files.
- Developed in-house by the Kerala High Court's IT department, the digital platform integrates Artificial Intelligence-based tools to generate structured case summaries and help judges quickly access digital records.
- CJI Surya Kant highlighted this initiative as a massive step toward "green jurisprudence," noting that transitioning to paperless operations in the ecologically sensitive Western Ghats will significantly lower the environmental footprint of the justice system.

Q 132. Ans. A

Explanation:

- **Statement 1 is incorrect:** The 'Creator's Corner' initiative was launched by Prasar Bharati (India's public service broadcaster), not the Press Information Bureau (PIB).
- **Statement 2 is correct:** It is a dedicated platform designed to recognize and empower India's growing community of digital content creators. It gives independent creators from across the country a chance to showcase their high-quality original reels and videos on national television, operating on a highly incentivized revenue-sharing model (90% of revenue to creators, 10% to Prasar Bharati).
- **Statement 3 is incorrect:** The content is specifically featured on DD News (telecast at 7:00 PM from Monday to Friday), not DD India.

Q 133. Ans. B

Explanation:

- The Challenger 150 Initiative is a global scientific cooperative endorsed by the UN Decade of Ocean Science for Sustainable Development (2021–2030). Its primary mission is to advance the mapping, understanding, and conservation of life in the deep ocean and seabed ecosystems across all ocean basins. It focuses on capacity building, standardizing deep-sea biological observations, and discovering life in underexplored regions.

Q 134. Ans. D

Explanation:

- According to the IQAir 2025 World Air Quality Report, Nieuwoudtville was identified as the world's cleanest city, recording an extraordinarily low annual average PM2.5 concentration of just 1.0 μg/m^3.
- India ranked as the 6th most polluted country globally (out of 143 nations monitored). The national average PM2.5 concentration stood at 48.9 μg/m^3, nearly 10 times the World Health Organization's safe limit of 5 μg/m^3.
- Loni, located in Uttar Pradesh, ranked as the single most polluted city in the world with an annual average PM2.5 concentration of 112.5 μg/m^3 (more than 22 times the WHO guideline).
- New Delhi retained its title as the most polluted capital city in the world for the eighth consecutive year.
- 66 of the world's 100 most polluted cities are located in India. In fact, three of the top four most polluted cities globally are Indian (Loni, Byrnihat, and Delhi).

Q 135. Ans. D

Explanation:

- Japanese mathematician Masaki Kashiwara was awarded the 2025 Abel Prize by the Norwegian Academy of Science and Letters. He made history by becoming the first Japanese national to receive this prestigious award, which is widely considered the mathematics equivalent of the Nobel Prize.
- The committee recognized his fundamental and pioneering contributions to algebraic analysis and representation theory spanning over half a century. Specifically, he was honored for his groundbreaking development of the theory of D-modules and the discovery of crystal bases, which have provided powerful new tools and opened entirely new avenues in modern mathematics.

Q 136. Ans. B

Explanation:

- Prime Minister Narendra Modi inaugurated the first edition of the SOUL (School of Ultimate Leadership) Leadership Conclave in February 2025 at Bharat Mandapam, New Delhi.
- The two-day conclave featured the Prime Minister of Bhutan, Dasho Tshering Tobgay, delivering the keynote address as the Guest of Honor.
- The School of Ultimate Leadership (SOUL) is an upcoming, first-of-its-kind leadership institution based in Gujarat. It aims to shape authentic, resilient, and conscientious leaders who can advance the public good across diverse domains, including politics, public policy, deep-tech, and the social sector.
- The initiative aims to broaden the landscape of leadership in India through formal training, enabling individuals who rise through merit, commitment, and passion for public service (rather than just political lineage) to navigate the complex challenges of the modern world.

Q 137. Ans. C

Explanation:

- Mark Carney, an economist and political newcomer, has been sworn in as Canada's new prime minister, and delivered remarks vowing to "never" become a part of the United States.
- Before entering politics, Carney was a renowned economist and served as the Governor of both the Bank of Canada and the Bank of England, making him the first person to head the central banks of two G7 countries

Q 138. Ans. C

Explanation:

- Bulgaria recently adopted the euro (€) as its official currency, becoming the 21st member of the euro area (Eurozone).
- Bulgaria fulfilled the required convergence criteria relating to inflation, public finances, exchange-rate stability, and long-term interest rates.
- The country replaced the Bulgarian lev (BGN) with the euro after receiving approval from the institutions of the European Union and the European Central Bank.

Q 139. Ans. D

Explanation:

- Kotak Mahindra Bank has recently launched India's first fully digital Foreign Portfolio Investor (FPI) licence, becoming the first custodian in the country to enable the entire FPI registration and account-opening process through electronic signatures (e-signatures) without requiring physical paperwork.
- The initiative was enabled by SEBI's unified digital workflow, operational since January 2026.
- The bank successfully issued the first FPI licences using completely digital documentation, eliminating the need for "wet signatures."

Q 140. Ans. B

Explanation:

- Mahendra Singh Dhoni has been appointed as the Goodwill Ambassador for the Pune Grand Tour 2026.
- The event, which took place in January 2026, marked India's first UCI-sanctioned international multi-stage professional road cycling race.

Q 141. Ans. A

Explanation:

Pattern: First number = Second number x (third number - 10)

34	26	102
17	13	17
12	12	16

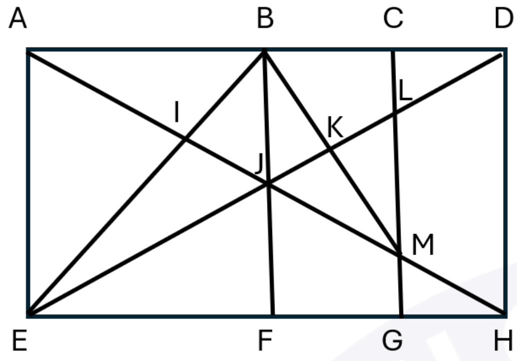
- **Column 1:**
  - ✓ Bottom = 12 →(12 − 10) = 2
  - ✓ Middle = 17
  - ✓ Top = 17 × 2 = 34
- **Column 2:**
  - ✓ Bottom = 12 →(12 − 10) = 2
  - ✓ Middle = 13
  - ✓ Top = 13 × 2 = 26
- **Column 3:**
  - ✓ Bottom = 16 →(16 − 10) = 6
  - ✓ Middle = 17
- Top = 17 × 6 = 102

Q 142. Ans. D

Explanation:

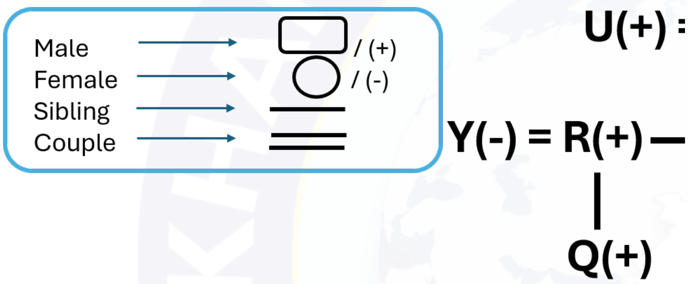
- **Triangles in the given diagram:** AIB, AEI, IEJ, AJE, EJF, JFM, MGH, JFH, AEH, BIJ, ABJ, BJK, JKM,

BMJ, ABM, BCM, KLM, JLM, LCD, BED, AJD, DJH, ADH.



Q 143. Ans. B

Explanation:



Q 144. Ans. C

Explanation:

Given:

- If + → −, − → ×, × → +
- **Expression:**
  - ✓ 240 + 15 − 6 × 4 + 8 × 3 − 10
- **Replace symbols:**
  - ✓ + → −, − → ×, × → +
- **So expression becomes:**
  - ✓ 240 − 15 × 6 + 4 − 8 + 3 × 10
- Apply BODMAS
- **Multiplication first:**
  - ✓ 15 × 6 = 90, 3 × 10 = 30
- **Now expression:**
  - ✓ 240 − 90 + 4 − 8 + 30
- Solve left to right
  - ✓ 240 − 90 = 150
    - 150 + 4 = 154
    - 154 − 8 = 146
    - 146 + 30 = 176

Q 145. Ans. D

Explanation:

Shifts:

- ✓ C → A = −2
- ✓ A → Y = −2
- ✓ N → K = −3
- ✓ D → A = −3
- ✓ L → H = −4
- ✓ E → A = −4

- **Pattern:** −2, −2, −3, −3, −4, −4
  - ✓ POCKET → NMZHAP

Q 146. Ans. B

Explanation:

- **Quadratic equation:** $16x^2 - kx + 25 = 0$
  - ✓ Here, a = 16, b = −k, c = 25
- **The condition for equal roots is:** $b^2 - 4ac = 0$
  - ✓ $(-k)^2 - 4 \times 16 \times 25 = 0$
  - ✓ $k^2 - 1600 = 0$
  - ✓ $k^2 = 1600$
  - ✓ $k = \pm 40$
- From the options,
  - ✓ k = 40

Q 147. Ans. A

Explanation:

$3\sqrt{3} \times 3^3 \div 3\text{-}1/2 = 3(x + 1)$

$3 \times 31/2 \times 3^3 \div 3\text{-}1/2 = 3(x + 1)$

$3(x + 1) = 3 \times 31/2 \times 3^3 \div 3\text{-}1/2$

$3(x + 1) = 3(1 + 1/2 + 3 + 1/2)$

$3(x + 1) = 3^5$

$x + 1 = 5$

$x = 5 - 1 = 4$

$\therefore 1/(1 + x^2) = 1/(1 + 4^2)$

$= 1/(1 + 16)$

$= 1/17$

Q 148. Ans. B

Explanation:

Weight of the new person = Weight of the old person + Increase in total weight

= 60 + 10 × 2 kg

= 60 + 20

= 80 kg

Weight of the new person = 80 kg

Q 149. Ans. C

Explanation:

- Probability of the male student being selected = 1/5
- Probability of the male student not being selected = 1 − 1/5 = 4/5
- Probability of the female student being selected = 1/3
- Probability of the female student not being selected = 1 − 1/3 = 2/3

- **Probability of only one being selected:**
  - ✓ $= (1/5 \times 2/3) + (1/3 \times 4/5)$
  - ✓ $= 2/15 + 4/15$
  - ✓ $= 6/15$
  - ✓ $= 2/5$

Ans. D

Explanation:

- **Ratio of the values of ₹1, 50 paise, and 25 paise coins = 4 : 3 : 2**
- $(4x \times 1) + (3x \times 2) + (2x \times 4) = 270$
- $4x + 6x + 8x = 270$
- $18x = 270$
- $x = 15$
- 50-paise coins = 6x
- $= 15 \times 6$
- $= 90$
- ∴ 50-paise coins = 90